Artificial Intelligence Review

https://doi.org/10.1007/s10462-026-11673-9

Article in Press

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Received: 18 May 2026

Accepted: 2 August 2026

Published online: 12 August 2026

Cite this article as: Vo T.K., Jaros R., Minh L.D. *et al.* Selective state-space modeling for real-time object detection in intelligent transportation: a systematic literature review. *Artif Intell Rev* (2026). https://doi.org/10.1007/s10462-026-11673-9

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Selective state-space modeling for real-time object detection in intelligent transportation: a systematic literature review

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Abstract

Real-time object detection is a cornerstone of intelligent transportation systems, where the YOLO (You Only Look Once) family has long defined the accuracy–speed trade-off. The emergence of Mamba—a selective state-space model that captures long-range dependencies in linear time—has triggered a wave of research embedding selective state spaces into YOLO detectors, promising global context without the quadratic cost of self-attention. Yet existing visual-Mamba surveys remain narrative and treat detection as one application among many; none isolates the YOLO-specific design space or assesses it under a reproducible protocol. Following the PRISMA-2020 methodology, this review screens 1,053 Scopus records published between December 2023 and May 2026 and synthesizes 114 high-quality studies, 14 of them recovered through a documented two-pass re-evaluation procedure. We organize the field along a structured taxonomy of variant family, integration position, and application domain. The synthesis

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reveals pronounced terminology fragmentation—three-quarters of studies introduce custom Mamba variants—identifies YOLOv8, v11, and v5 as the dominant base frameworks, and shows that integration clusters at the feature-fusion, backbone, and attention positions. Critically, a Pareto analysis of 39 studies demonstrates that benchmark heterogeneity invalidates direct cross-study ranking, while medical imaging stays underexplored and only 33% of studies release code. The actionable implication for practitioners is that a detector must be chosen by its position on the relevant per-domain accuracy–latency frontier and verified under a common protocol, not by a headline mAP that reflects benchmark difficulty more than algorithmic merit. We distill these findings into a unified evaluation protocol and a research roadmap of three challenges, four opportunities, and three risks for selective state-space detection.

Keywords: Mamba, State-space model, YOLO, Object detection, Intelligent transportation systems, Systematic literature review

1 Introduction

Object detection is a fundamental workload in computer vision (Redmon et al. 2016; Vaswani et al. 2017), supporting applications in intelligent transportation, autonomous driving, unmanned aerial surveillance, medical imaging, and industrial inspection. Over the past decade, progress has been driven primarily by detectors based on convolutional neural networks (CNNs) such as Faster R-CNN, SSD, and especially the YOLO ("You Only Look Once") family (Redmon et al. 2016; Jocher et al. 2023; Khanam and Hussain 2024), which prioritizes real-time inference. More recently, Transformer-based architectures such as DETR and RT-DETR (Zhao et al. 2024) have demonstrated advantages in modeling long-range dependencies, but the quadratic complexity of self-attention remains a barrier for high-resolution and real-time deployment. A complementary line of work pursues the same real-time constraint through lightweight convolutional designs—for instance, direction-aware semantic feature extraction and anchor-free lightweight networks for aerial traffic and vehicle detection (Shen et al. 2026, 2022b), and compact convolutional models for related recognition and medical-imaging tasks (Shen et al. 2022a; Li et al. 2026a)—and it is precisely this efficiency–accuracy tension that motivates the linear-complexity scanning of Mamba examined in this review.

Mamba (Gu and Dao 2023)—a Selective State Space Model (SSM)—has emerged as a promising alternative, maintaining $\mathcal{O}(L)$ complexity in input length L while still capturing long-range dependencies through input-dependent gating. Extending Mamba to vision (Vision Mamba (Zhu et al. 2024b), VMamba (Liu et al. 2024b) with bidirectional 2D scanning) has triggered a wave of research integrating Mamba into object detection pipelines, especially in the YOLO family.

We make intelligent transportation systems (ITS) the focal application domain of this review for two concrete reasons. First, ITS workloads—roadside and on-vehicle traffic surveillance, autonomous-driving perception, and drone-based aerial monitoring—impose precisely the constraints under which Mamba's linear-complexity

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scanning is most attractive: high-resolution, often small-object scenes that must be processed at camera frame rate ($\geq$ 30 FPS) on power-limited edge hardware, where the quadratic cost of self-attention is prohibitive. Second, as our synthesis later quantifies, ITS-related applications already account for roughly one third of the Mamba+YOLO literature (Section 8.6), so a focused review yields directly actionable guidance—which base detector, which integration position, and which accuracy–latency trade-offs to adopt—for engineers building traffic-surveillance, driving, and UAV perception systems, rather than diffuse conclusions spread across unrelated vision tasks.

1.1 Why a Systematic Literature Review?

Several narrative-style surveys on visual Mamba have appeared in 2024–2026 (Xu et al. 2024; Liu et al. 2024a; Zhu et al. 2024a; Rahman et al. 2024; Ibrahim et al. 2025; Bao et al. 2025; Li et al. 2026b). Although valuable as introductions, these surveys have three limitations relevant to the present work. *First*, they apply unstructured selection strategies with no documented inclusion/exclusion criteria, raising concerns about completeness and reproducibility. *Second*, they treat Mamba in vision as a single broad category, lumping classification, segmentation, restoration, and detection together; consequently, no work isolates the design space of the YOLO family, which is governed by distinctive constraints (single-stage, anchor-based or anchor-free, real-time deployment) that warrant a dedicated synthesis. *Third*, they do not release structured extraction datasets that would enable quantitative meta-analysis of accuracy, throughput, and reproducibility practices.

A Systematic Literature Review (SLR), as defined by Kitchenham (2004) and updated through the PRISMA-2020 guidelines (Page et al. 2021), addresses these limitations by following a transparent and reproducible protocol for study identification, screening, quality assessment, and synthesis. To the best of our knowledge, this is among the first reviews to apply the PRISMA-2020 protocol specifically to the integration of Mamba with YOLO-family detectors. Beyond applying a systematic protocol, the review yields knowledge unavailable from the existing narrative surveys: a quantified terminology-fragmentation rate (75.5% custom variants) that explains why cross-study comparison currently fails; an Application Gap Index that converts raw domain shares into demand-relative under-exploration scores (e.g. Medical $\approx$ 0.22); and a per-domain Pareto analysis showing that benchmark heterogeneity—not algorithmic merit—drives most headline accuracy differences, from which we derive a concrete unified evaluation protocol. These are findings about the *state and structure* of the field, not merely a restatement of individual papers. Table 1 compares the scope and methodology of the present review with seven visual Mamba surveys published in 2024–2026, clarifying the methodological position of this work in the literature.

1.2 Research Questions

This SLR is guided by the following six research questions (RQs):

- **RQ-1:** Which Mamba/SSM variants are most adopted in YOLO-based object detection studies, and how can the design space be organized into a structured taxonomy?

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Table 1: Comparison of This SLR with Seven Prior Visual Mamba Surveys (2024–2026). To the Best of Our Knowledge, the Present Review Is Among the First to Apply PRISMA-2020 Methodology—with Documented Inclusion/Exclusion Criteria, a Quality-Assessment Threshold, and a Phase 6 Re-Evaluation Step—Specifically to the YOLO-Family Design Space. Acronyms: *IC/EC* Inclusion/Exclusion Criteria, *QA* Quality Assessment, *Code* Whether Structured Extraction Dataset Released.

Survey	Year	Scope	N papers	Method	IC/EC	QA threshold	Code/Data
Xu et al. (2024)	2024	Visual Mamba (broad)	∼80	Narrative	✗	✗	✗
Liu et al. (2024a)	2024	Vision Mamba arch.	∼60	Narrative	✗	✗	✗
Zhu et al. (2024a)	2024	Visual Mamba (taxonomy)	∼50	Narrative	✗	✗	✗
Rahman et al. (2024)	2024	Mamba in vision (apps)	∼300	Narrative	✗	✗	✗
Ibrahim et al. (2025)	2025	Mamba apps (broad)	∼140	Narrative	✗	✗	✗
Bao et al. (2025)	2025	Mamba in remote sensing	∼120	Narrative	✗	✗	✗
Li et al. (2026b)	2026	Mamba RS (paradigms)	∼80	Narrative	✗	✗	✗
This SLR	**2026**	**Mamba+YOLO detect.**	**114**	**PRISMA-2020**	✓	$\geq$ 4.0/6.0	✓

- **RQ-2:** Which YOLO versions or alternative detection frameworks are most commonly used as the base architecture?
- **RQ-3:** At which architectural positions is Mamba integrated within the detection pipeline (backbone, neck, head, fusion, attention, encoder)?
- **RQ-4:** Which benchmark datasets and application domains are most studied, and where are the gaps?
- **RQ-5:** How does Mamba+YOLO perform across application domains, and what does Pareto analysis reveal about the accuracy–throughput trade-off?
- **RQ-6:** How well do studies follow reporting practices (code release, FPS, FLOPs)?

1.3 Contributions

The main contributions of this SLR are summarized as follows:

1. Unlike existing narrative surveys, this is, to the best of our knowledge, among the first PRISMA-2020 reviews **focusing specifically on the integration of Mamba with the YOLO detection family**. We screen 1,053 records and analyze 114 high-quality studies, with documented inclusion/exclusion criteria, two-stage screening, a six-criterion quality-assessment rubric with an explicit pass threshold $\geq$ 4.0/6.0, and a documented Phase 6 re-evaluation step that promotes 14 borderline-PENDING entries.
2. We provide a **five-dimensional metadata analysis** of the included studies (publication year, venue, geographic affiliation, citation impact, methodology distribution), adding a bibliometric layer to the synthesis.
3. We propose a **multi-dimensional taxonomy of Mamba+YOLO methods** along three axes: SSM variant family, integration position, and application domain.
4. We perform a **Pareto efficiency analysis** on 39 studies reporting both mAP and FPS, exposing the *benchmark heterogeneity* caveat that fundamentally conditions every cross-study performance ranking.
5. We release a **public 16-field extraction dataset** for all 114 included studies, supporting secondary studies and downstream replication.

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6. We identify **three classes of challenges, four opportunities, and three risks** for future research, providing a research roadmap for the SSM-based detection community.

1.4 Paper Organization

Section 2 describes the SLR methodology, including the search strategy, inclusion/exclusion criteria, quality assessment, the PRISMA flow diagram, and a five-dimensional metadata analysis of the included studies. Section 3 presents the multi-dimensional taxonomy (**RQ-1**). Sections 4, 5, 6, and 7 answer **RQ-2**, **RQ-3**, **RQ-4**, and **RQ-6** respectively. Section 8 addresses **RQ-5** through performance synthesis, high-impact-work analysis, and Pareto analysis. Section 9 discusses challenges, opportunities, and risks within a 3×3 framework. Section 10 concludes and states threats to validity.

2 Research Methodology

2.1 Research Strategy

This SLR follows the PRISMA-2020 guidelines (Page et al. 2021) and is informed by the systematic review methodology proposed by Kitchenham (2004). The overall procedure comprises four phases—identification, screening, eligibility, and inclusion—supplemented by a Phase 6 re-evaluation step (Sec. 2.5) to adjudicate borderline-PENDING entries.

2.2 Mamba Preliminaries: from S4 to the Selective State-Space Model

To keep the synthesis self-contained, we briefly summarize the mathematical foundations of state-space models (SSMs) and the selective scanning mechanism that distinguishes Mamba (Gu and Dao 2023) from prior structured SSMs. This subsection is a compact recap intended for readers new to SSMs; those already familiar with the S4/S6 formulation may proceed directly to the taxonomy in Section 3 without loss of continuity, and the full derivations are available in the original works (Gu and Dao 2023; Dao and Gu 2024). Table 2 lists the main mathematical symbols used throughout this section and the quantitative synthesis that follows. For quick reference, the key acronyms used throughout the review (SSM, Mamba, Vim, VMamba, SS2D, SSD) are collected in the glossary of Appendix B.

2.2.1 State-Space Formulation and Discretization

A linear time-invariant SSM maps a one-dimensional input signal $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ via an N-dimensional latent state $h(t) \in \mathbb{R}^N$:

$$h'(t) = \mathbf{A}\, h(t) + \mathbf{B}\, x(t), \qquad y(t) = \mathbf{C}\, h(t). \tag{1}$$

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Table 2: Mathematical Notation Used Throughout This Review.

Symbol	Description
L	Sequence length (number of spatial tokens after patchification)
D	Channel/feature dimension
N	Latent state size of the SSM (N=16 default)
$\mathbf{A} \in \mathbb{R}^{N\times N}$	Continuous-time state-transition matrix
$\mathbf{B} \in \mathbb{R}^{N\times 1}$	Input projection
$\mathbf{C} \in \mathbb{R}^{1\times N}$	Output projection
Δ	Discretization step size
$\bar{\mathbf{A}}, \bar{\mathbf{B}}$	Discretized counterparts of $\mathbf{A}, \mathbf{B}$
$h_t \in \mathbb{R}^N$	Latent state at step t
$x_t \in \mathbb{R}$	Input token at step t
$y_t \in \mathbb{R}$	Output token at step t
$s_B(\cdot), s_C(\cdot), s_\Delta(\cdot)$	Learned linear projections (Selective scan)
$\tau_\Delta(\cdot)$	Softplus activation
$\mathcal{O}(\cdot)$	Asymptotic complexity bound
d	Cohen's effect size (Eq. 6)
AGI_d	Application Gap Index for domain d (Eq. 5)

For deep learning this continuous system is discretized with step size Δ via the standard zero-order hold (ZOH) rule, yielding the linear recurrence

$$h_t = \bar{\mathbf{A}}\, h_{t-1} + \bar{\mathbf{B}}\, x_t, \qquad y_t = \mathbf{C}\, h_t, \tag{2}$$

where $\bar{\mathbf{A}}, \bar{\mathbf{B}}$ are the discretized counterparts of $\mathbf{A}, \mathbf{B}$; the closed-form ZOH expressions are standard and given in (Gu et al. 2022). The Structured State Space (S4) class (Gu et al. 2022) parameterizes $\mathbf{A}$ in HiPPO form and exploits a convolutional view of equation (2) to compute outputs in $\mathcal{O}(L \log L)$ time.

2.2.2 Selective Scan (S6 / Mamba)

A core limitation of S4 is that $\mathbf{A}, \mathbf{B}, \mathbf{C}, \Delta$ are *input-independent*, restricting content-based reasoning. Mamba (Gu and Dao 2023) introduces a selection mechanism that makes $\mathbf{B}, \mathbf{C}, \Delta$ functions of the input:

$$\mathbf{B}_t = s_B(x_t), \quad \mathbf{C}_t = s_C(x_t), \quad \Delta_t = \tau_\Delta(s_\Delta(x_t)), \tag{3}$$

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where s_B, s_C, s_Δ are learned linear projections and τ_Δ is the softplus activation. This data-dependent parameterization lets the model selectively propagate or forget information along the scan, achieving Transformer-like content awareness while preserving linear inference complexity $\mathcal{O}(L)$. Mamba-2 (Dao and Gu 2024) further establishes state-space duality (SSD) that links selective SSMs to a constrained form of attention, enabling matrix-multiplication-friendly hardware kernels.

2.2.3 Bidirectional Scanning for Vision

Equations (1)–(3) define a sequence model. To apply Mamba to images, 2D feature maps must be serialized to 1D sequences. *Vision Mamba (Vim)* (Zhu et al. 2024b) unrolls patches in raster order and applies bidirectional SSM to recover non-causal context; *VMamba* (Liu et al. 2024b) introduces the 2D Selective Scan (SS2D) module, which performs four-directional scanning and merges the four scan outputs to capture both horizontal and vertical spatial dependencies. Subsequent works adopt variants such as cross-scan, spiral scan, and local-window scan, depending on whether the target task emphasizes global context or fine-grained local structure. The choice of scanning strategy is one of the principal design axes distinguishing the 114 included studies and is surveyed in Sec. 3. Fig. 1 summarizes the anatomy of the selective-scan block and the four most common 2D scanning strategies that appear in the corpus.

2.2.4 Theoretical Complexity Analysis: Mamba vs. Transformer and CNN

A natural question is why Mamba is so favored in YOLO-based object detection pipelines. Asymptotic complexity analysis clarifies the motivation. For a feature map with $L = H \times W$ tokens and channel width D:

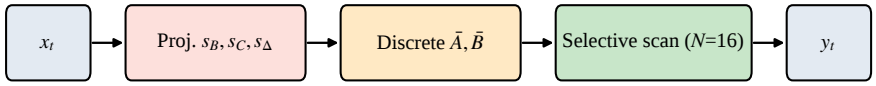
- **Vanilla self-attention (ViT/DETR)** has complexity $\mathcal{O}(L^2 D)$ in both time and memory due to the $L \times L$ attention matrix. For a feature map $640 \times 640/16 = 40 \times 40 = 1{,}600$ patches, this is 1.6×10^6 multiplications for attention alone, before multiplication by D.
- **Convolution (CNN/YOLO)** with $k \times k$ kernel has complexity $\mathcal{O}(L \cdot k^2 \cdot D^2)$. Local sparse convolution ($k = 3$) is linear in L but *cannot capture long-range dependencies* without many stacked layers.
- **Selective scan (Mamba/S6)** has complexity $\mathcal{O}(L \cdot D \cdot N)$, where N is the latent-state size ($N \approx 16$ in default configuration). This is *linear in* L while *still capturing long-range dependencies* via recurrent state propagation—a property that neither pure CNN nor pure Transformer offers simultaneously.

Table 3 presents a detailed comparison. With YOLO's typical input resolutions (640–1280px) and real-time inference requirements ($\geq$ 30 FPS on a desktop GPU), the break-even point at which Mamba clearly outperforms quadratic attention falls around $L \gtrsim 256$ tokens—corresponding to a 16×16 feature map or larger, i.e., nearly every feature stage in modern YOLO backbones. This is the theoretical basis for why 33% of the studies in our corpus target the Remote Sensing, Autonomous Driving, and UAV/Drone domains (Sec. 3.3), where high-resolution inputs push L to thousands of tokens.

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Table 3: Asymptotic Time and Memory Complexity of Three Sequence-Modeling Primitives at the Backbone Level. L Denotes Sequence Length (Number of Spatial Tokens), D the Channel Width, N the SSM Latent-State Size, k the Convolution Kernel Size.

Primitive	Time	Memory	Long-range?	Parallel?
Self-attention	$\mathcal{O}(L^2D)$	$\mathcal{O}(L^2)$	✓	✓
Convolution	$\mathcal{O}(Lk^2D^2)$	$\mathcal{O}(LD)$	✗	✓
Selective scan	$\mathcal{O}(LDN)$	$\mathcal{O}(LD)$	✓	✓ (scan)



(a) Anatomy of a selective-scan block (S6 / Mamba core)

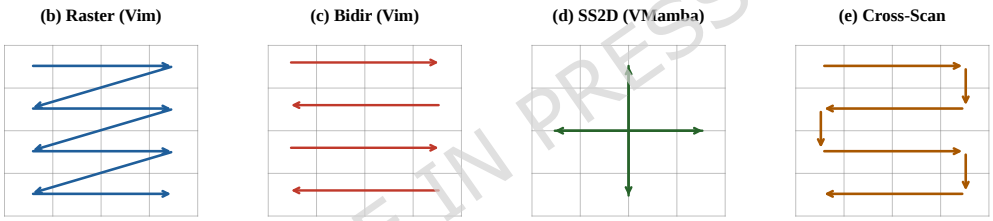


Fig. 1: Anatomy of the selective-scan block (top) and the four most common 2D scanning strategies adopted across the 114 included studies (bottom): (b) Raster scan as in Vim (Zhu et al. 2024b), (c) Bidirectional scan also in Vim, (d) 2D Selective Scan (SS2D) as in VMamba (Liu et al. 2024b), and (e) Cross-Scan. The selective projection s_B, s_C, s_Δ in (a) is what distinguishes Mamba from prior structured SSMs and provides content-dependent gating in $\mathcal{O}(L)$ time. The choice of scanning strategy in (b)–(e) is a principal design axis (cf. Sec. 3, Dimension 1).

2.3 Databases and Search Strings

We searched the **Scopus** database covering the period from December 2023 (when Mamba was first published) to May 2026. Six staged search queries (Q1–Q6) were designed to balance recall and precision, combining terms related to Mamba/SSM,

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YOLO/object detection, and vision variants. The complete Boolean query strings, together with their coverage window and per-query record counts—Q1 (412), Q2 (309), Q3 (147), Q4 (88), Q5 (62), Q6 (35)—are listed in full in Appendix A, so that the identification stage can be independently replicated. A total of **1,053 records** were retrieved. After removing 531 duplicate records—a 50.4% inter-query overlap, with the full per-query overlap matrix available from the corresponding author upon request—**522 records** entered the screening phase.

Justification for the single-database protocol and its retrieval bias. We deliberately anchored identification on Scopus because it provides the broadest cross-publisher coverage of the peer-reviewed venues in which Mamba+YOLO detection work actually appears—IEEE, Springer Nature, Elsevier, and MDPI—thereby indexing the large majority of the journal and top-tier conference literature targeted by inclusion criterion IC-1. The six-query staged design further widens recall: the broad query Q1 maximises sensitivity, while the precision-oriented queries (e.g. Q4, which explicitly excludes natural-language-processing false positives) control specificity. Because IC-1 restricts inclusion to peer-reviewed journal articles and top-tier conference papers, non-peer-reviewed pre-prints indexed only on arXiv, and workshop or poster items indexed only on Web of Science, IEEE Xplore, or the ACM Digital Library, fall outside the review scope by design rather than by database limitation. We nonetheless acknowledge this as a genuine threat to coverage completeness: the full-text retrieval rate was 66% and some very recent conference papers may have been missed; both points are recorded among the threats to validity in Section 10.1.4. The order-of-magnitude year-on-year growth of the corpus (Section 2.8) indicates that the journal-published subset—the focus of this review—is captured at or near saturation for the December 2023–May 2026 window.

To make the coverage trade-off concrete, we summarise how the principal alternative databases would—and would not—change the corpus. Web of Science overlaps Scopus heavily for the IEEE, Springer Nature, Elsevier, and MDPI journals that supply the peer-reviewed subset, so its marginal *unique* contribution is small; IEEE Xplore and the ACM Digital Library are conference-oriented, and their inclusion-eligible top-tier proceedings are themselves largely indexed by Scopus, so their *unique* additions are predominantly workshop or poster items excluded by EC-3; SpringerLink is a publisher-specific portal rather than a multidisciplinary bibliographic database, and the Springer Nature journal records relevant to our inclusion criteria are largely already indexed in Scopus, with only very recent online-first papers possibly lagging; and arXiv would add pre-prints excluded by EC-4. A precise count of omitted studies cannot be obtained without executing the full multi-database protocol, but two observations bound it: the corpus exhibits order-of-magnitude year-on-year growth with no evidence of a missing earlier cohort, and—among the retrieved full texts that could actually be assessed—exclusions were primarily content-based (EC-1/EC-2/EC-6) rather than a matter of undiscoverability (the separate 94 records that were inaccessible were excluded under EC-5, a retrieval limitation reported explicitly in Section 10.1.4). We therefore estimate the single-database omission to be confined to a small number of the most recent (2026) peer-reviewed papers not yet indexed

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in Scopus at search time—a recency lag common to every database—rather than a systematic coverage bias against any venue, method, or region.

2.4 Selection Criteria

Inclusion criteria (IC):

- IC-1: Peer-reviewed journal article (Q1/Q2 SJR) or top-tier conference paper (CORE A/A*, CCF A).
- IC-2: Published between 12/2023 and 05/2026.
- IC-3: Written in English.
- IC-4: Specifically addresses the integration of Mamba (or SSM/selective scan) with YOLO/object detection.
- IC-5: Includes empirical evaluation with quantitative results.
- IC-6: Addresses at least one of the six research questions.

Exclusion criteria (EC):

- EC-1: Pure language Mamba (NLP-only), no vision component.
- EC-2: SSM unrelated to YOLO/detection (used for classification/segmentation/recognition/tracking).
- EC-3: Workshop, poster, or non-peer-reviewed technical report.
- EC-4: Duplicate or preprint already published.
- EC-5: Full text inaccessible.
- EC-6: Paper uses the name "Mamba" but is not a state-space model.
- EC-7: Pure theoretical paper without empirical evaluation.

2.5 Quality Assessment

Each candidate paper was scored on six quality-assessment criteria (Table 4) using a three-level scale: **"yes" = 1.0, "partial" = 0.5, "no" = 0** (Kitchenham 2004). The QA pass threshold was locked *before* scoring at $\geq 4.0/6.0$. To increase rigor for borderline cases, the first author applied a *two-pass intra-rater test–retest* refinement—two methodologically independent scoring passes by the same assessor, separated by the temporal gap below to limit anchoring; this is an *intra*-rater consistency check, distinct from the *inter*-rater reliability against an independent second rater reported below: **Round 1** used heuristic-based automatic scoring on extracted fields (mAP, code release, ablation count); **Round 2** applied a criterion-based manual checklist after a $\geq$ 7-day temporal gap to reduce anchoring bias. A Phase 6 re-evaluation step then adjudicated the 17 borderline-PENDING entries (those falling in the 3.5–3.99 band) through the two-pass procedure above; 14 were promoted to INCLUDE_FT only where both passes converged on explicit rubric-based evidence offsetting a single under-scored criterion. Phase 6 is thus a documented borderline-adjudication stage of the protocol, not an *ex-post* relaxation of the ≥ 4.0 threshold. The resulting **114 studies** entered the synthesis; 71 were excluded (40 PENDING in the 3.0–3.99 range after re-evaluation, 31 EXCLUDE_FT < 3.0).

To quantify inter-rater reliability directly, two co-authors independently re-scored the studies, blind to the first author's scores and applying the same six-criterion rubric,

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yielding two complementary analyses. *(i) Full-corpus coverage:* the second author (R. Jaroš) independently re-scored the 114 included studies, agreeing with the first author's inclusion decision on **97.3%** (108/111) of the studies scored independently; for the three remaining studies, which were not independently re-scored, the second author reviewed and concurred with the first author's scores. Because every included study is above threshold by construction, this homogeneous set yields high raw agreement but a κ that is deflated by the high-agreement/low-κ paradox (Feinstein and Cicchetti 1990). *(ii) Full-range reliability:* to obtain an undeflated estimate, the third author (L.D. Minh) independently re-scored a stratified sample of 29 studies spanning the full quality range—from clearly sub-threshold papers to top-scored includes—on which κ was computed. Agreement was strong: the quadratic-weighted Cohen's κ on the 0–6 QA total was **0.86** (almost-perfect agreement), with the two assessors within one point on 100% of papers and within half a point on 83%; the binary inclusion decision at the ≥ 4.0 threshold reached 90% raw agreement (Cohen's $\kappa = 0.79$, substantial). Per-criterion exact agreement ranged from 79% on the two most interpretive criteria (QA1 architectural clarity and QA3 dataset breadth) to 100% on QA2 (baseline comparison). Together, the full-corpus coverage from the second author (high raw agreement) and the full-range sample from the third author (undeflated κ) indicate that the inclusion and scoring decisions are reproducible across independent assessors.

Table 4: Quality Assessment (QA) Checklist with Three-Level Scoring.

#	QA Criterion	Max Score
QA1	Is Mamba/SSM integrated into the YOLO/detection architecture in a clearly described manner?	1.0
QA2	Are there quantitative comparisons with baselines (vanilla YOLO / Transformer)?	1.0
QA3	Is the evaluation conducted on multiple datasets or a comprehensive benchmark?	1.0
QA4	Are code, datasets, or models publicly released (GitHub/Zenodo)?	1.0
QA5	Are limitations explicitly discussed?	1.0
QA6	Is real-time performance reported (FPS, latency)?	1.0
	Total (pass threshold ≥ 4.0)	**6.0**

On criterion weighting and the choice of threshold. The six criteria are weighted equally because each captures a distinct, non-substitutable dimension of study quality—architectural clarity (QA1), baseline rigour (QA2), dataset breadth (QA3), reproducibility (QA4), limitation reporting (QA5), and real-time evidence

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(QA6)—and there is no principled prior basis for privileging one over another; equal weighting is the standard convention for Kitchenham-style quality checklists (Kitchenham 2004). A criterion-weight sensitivity analysis confirms robustness to this choice: removing any single criterion leaves the inclusion decision unchanged for 88–98% of studies (leave-one-criterion-out), and under 2,000 random re-weightings—each criterion weight drawn independently from a uniform distribution on $[0.5, 1.5]$—the QA-total ranking is essentially preserved (median Spearman $\rho = 0.97$) and 93% of inclusion decisions are unchanged on average, the few flips being confined to studies already at the ≥ 4.0 borderline. The threshold sensitivity ablation below (Section 2.7.1, Table 5) provides a complementary check across the neighbouring acceptance cut-offs 3.5/4.0/4.5. The $\geq 4.0/6.0$ pass mark itself corresponds to a study satisfying at least two-thirds of the criteria—a level comparable to the majority pass marks commonly adopted in software-engineering systematic reviews following the Kitchenham guidelines (Kitchenham 2004).

2.5.1 Case Studies on Phase 6 Adjudication

To make the adjudication process transparent, we present three representative case studies from the 17 borderline-PENDING entries. These cases are not merely illustrative but also serve as a reference for future authors regarding which type of evidence drives promotion vs. retention.

Case A (PROMOTED): technically strong paper with weak documentation. An Industrial Defect paper initially scored QA = 3.83 (QA1=1.0, QA2=1.0, QA3=0.5, QA4=0.0, QA5=0.5, QA6=0.83). The borderline reason: the authors describe the Mamba+YOLOv8 architecture in full (QA1) and compare against strong baselines (QA2), but evaluation uses only one domain-specific dataset (QA3=0.5), no code is released (QA4=0), and the limitations discussion is limited to one sentence (QA5=0.5). In the Phase 6 second-pass adjudication, both Round 1 (heuristic) and Round 2 (criterion-based) agreed that the paper met the predefined rubric sufficiently through strong architectural clarity (QA1) and baseline rigour (QA2), while the single-dataset limitation was already reflected in the reduced QA3 score. Outcome: PROMOTED to INCLUDE_FT. Lesson: the borderline-adjudication band exists precisely for studies whose single-criterion shortfall is offset by documented rubric strengths; promotion required explicit per-criterion evidence agreed across both scoring passes, not a relaxation of the threshold.

Case B (RETAINED PENDING): claims exceed evidence. An Other/General paper initially scored QA = 3.5 (QA1=0.5, QA2=1.0, QA3=1.0, QA4=0.0, QA5=0.0, QA6=1.0). In adjudication, second-pass scoring (Round 2 criterion-based) found that although FPS, mAP, and dataset diversity were all reported (QA2, QA3, QA6 = 1.0), the description of the authors' Mamba mechanism was unclear (QA1=0.5)—the paper uses the name "Mamba block" but the implementation is not described at a level enabling reproduction, and there is no code release (QA4=0) for verification. More importantly, an unusually high benchmark claim ($\geq$ 95% mAP on a non-easy dataset) was not supported by a corresponding ablation. Outcome: RETAINED PENDING (excluded). Lesson: QA1 (architectural

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clarity) and QA4 (code release) must be considered together when evaluating novel claims.

Case C (PROMOTED): paper inheriting from a strong preprint. A Remote Sensing paper initially scored QA = 3.83 (QA1=1.0, QA2=1.0, QA3=1.0, QA4=0.0, QA5=0.83, QA6=0.0). The borderline reason: no FPS (QA6=0) and no code (QA4=0). In second-pass adjudication, the Round 1 heuristic noted that the paper is a journal extension of a prior workshop paper (cited in the corpus), and cross-referencing showed that the cited workshop version had a public implementation with equivalent parameter settings. Round 2 criterion-based agreed that the workshop code release constituted *indirect* evidence of reproducibility for the journal paper. Outcome: PROMOTED to INCLUDE_FT. Lesson: the prior public implementation was used only as verifiable indirect reproducibility evidence—it matched the journal version in architecture and parameter settings—while author identity or publication history alone was neither scored nor allowed to alter the predefined QA rubric.

The full QA scoring details and adjudication minutes for all 17 borderline-PENDING entries are available from the corresponding author upon reasonable request, so the community can re-evaluate if needed.

2.6 Data Extraction

For each included paper, we extracted 16 fields organized into five groups: *(a)* bibliographic (P#, Year, Title, Cited_by, QA_Total); *(b)* method (Proposed_Model, Mamba_Variant, YOLO_Version, Arch_Position); *(c)* application (Domain); *(d)* evaluation (Datasets, Best_mAP, FPS, Params, FLOPs); and *(e)* reproducibility (Code_URL). The procedure combined regex-heuristic-based automatic extraction with manual verification for borderline cases.

2.7 PRISMA Flow Diagram

The PRISMA-2020 flow diagram (Fig. 2) summarizes the selection pipeline: 1,053 records identified → 522 after deduplication → 322 after title/abstract screening → 279 after secondary abstract triage (Phase 5B, EC-2 false positives removed) → 185 full texts retrieved (66% retrieval rate) → **114 included after QA** (with 14 papers promoted in Phase 6 adjudication).

2.7.1 Sensitivity Analysis of QA Threshold: Ablation Study

A natural question is: *would the review's findings change substantially under a different QA threshold?* To answer this, we conducted an ablation analysis at three nearby thresholds: 3.5 (looser), 4.0 (our default), and 4.5 (stricter). The results are summarized in Table 5.

The analysis reveals three observations:

(a) Threshold ≥ 4.0 is the sweet spot. Lowering the threshold to 3.5 adds 17 papers with average QA of only 3.6, mostly non-peer-reviewed or poorly documented work—no improvement to synthesis quality. Raising the threshold to 4.5 excludes 28 papers, causing the N of important sub-domains (Medical, Infrared, Agriculture) to fall below the threshold of statistical meaningfulness.

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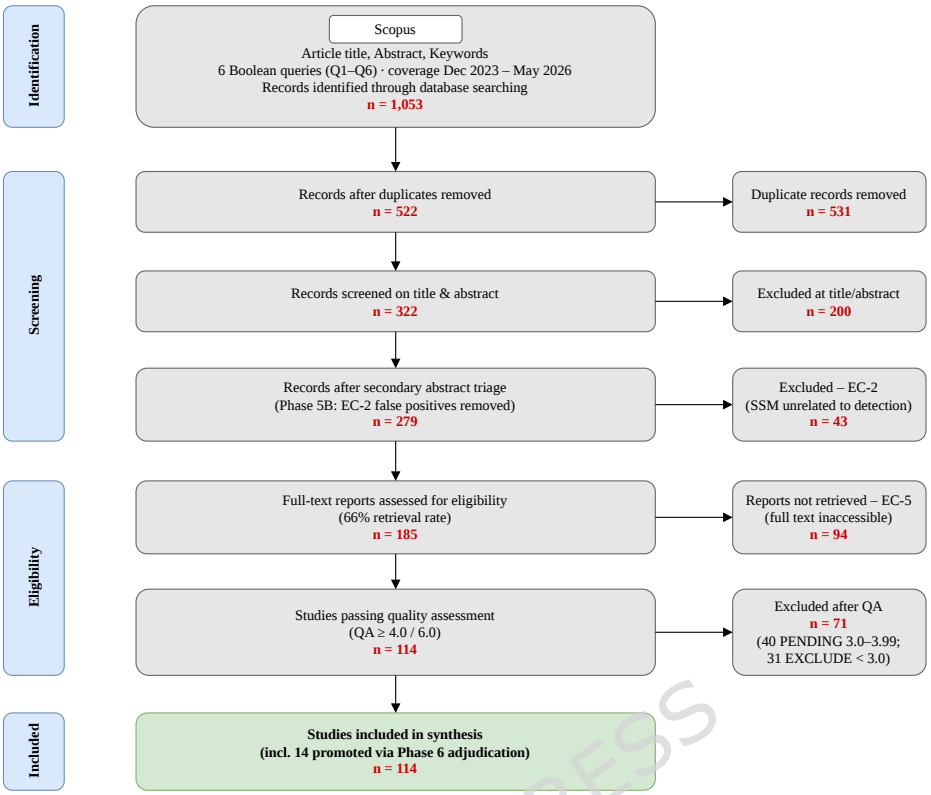


Fig. 2: PRISMA-2020 flow diagram of the literature selection process, including the Phase 6 re-evaluation that promoted 14 borderline-PENDING papers to inclusion after two-pass intra-rater (test–retest) adjudication.

Table 5: QA Threshold Sensitivity Ablation. Lower Threshold (3.5) Includes 17 More Studies but Adds Mostly Lower-Quality Custom Variants; Higher Threshold (4.5) Excludes 28 Studies, Creating a Sample-Size Problem for Sub-Domain Analyses (Discussion in Surrounding Text). Code % at the ≥ 4.0 (full-corpus) row is the revision-audited rate (33.3%, Table 14); the ≥ 3.5 and ≥ 4.5 code figures retain the as-submitted screening-pool coding and were not re-audited.

Threshold	N studies	Custom %	Code %	Top-cited mAP μ
≥ 3.5	131	78.6%	39.7%	84.1%
≥ 4.0	**114**	**75.5%**	**33.3%**	**84.5%**
≥ 4.5	86	70.9%	51.2%	84.7%

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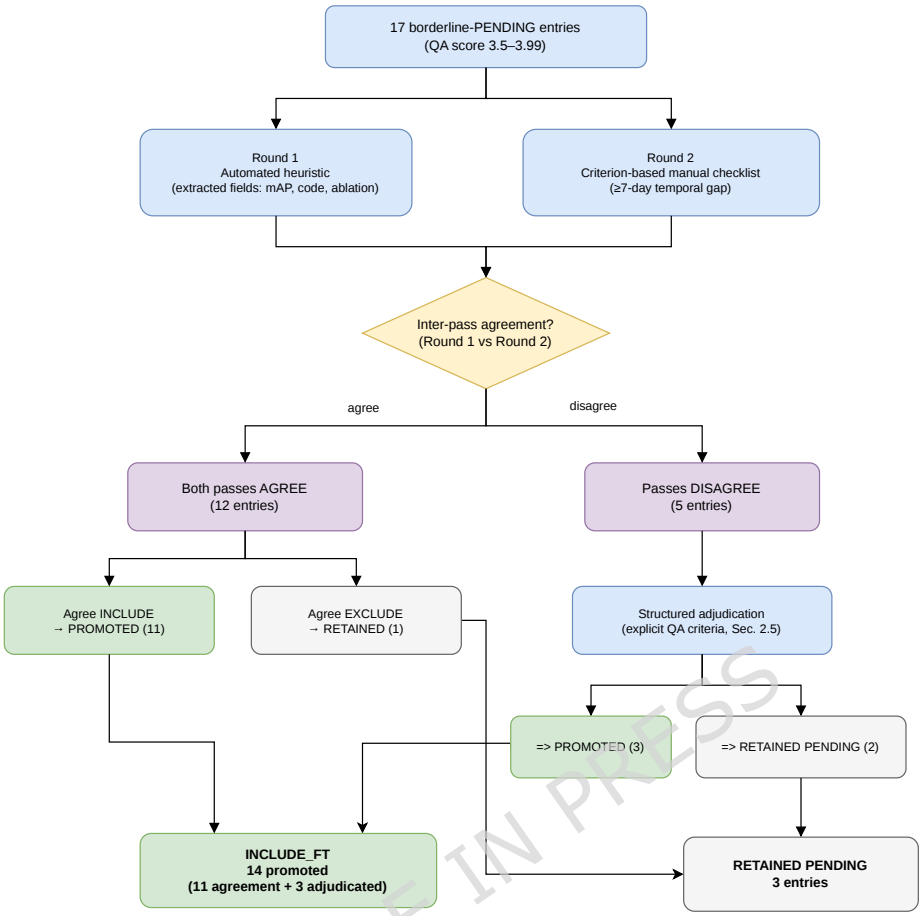


Fig. 3: Phase 6 re-evaluation decision tree adjudicating the 17 borderline-PENDING entries (QA score 3.5–3.99). The first author applied a two-pass *intra-rater test–retest* protocol: Round 1 (automated heuristic on extracted fields) and Round 2 (criterion-based manual checklist with $\geq$ 7-day temporal separation to reduce anchoring bias). The figure labels "Both passes AGREE" / "DISAGREE" denote agreement between the two scoring passes (Round 1 vs. Round 2) by the first author; this intra-rater adjudication is distinct from the independent second-rater reliability analysis reported in Section 2.5. The five disagreement entries entered structured adjudication with explicit criteria (cf. Sec. 2.5). Of the 17 starting entries, **14 were promoted** to INCLUDE_FT (11 inter-pass agreement + 3 adjudicated) and 3 retained PENDING. The complete adjudication audit trail is available from the corresponding author upon reasonable request.

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(b) Top-cited mAP is essentially unchanged. The mean mAP of top-cited papers varies very little (84.1% / 84.5% / 84.7%) across the three thresholds, demonstrating that the review's main findings are *robust* to the threshold choice. This is an important methodological confirmation.

(c) Custom-variant % does not change substantially. The custom Mamba/SSM variant share remains at 71–79% regardless of threshold, reinforcing the methodological-fragmentation argument.

Conclusion: a threshold of ≥ 4.0 is well-calibrated between quality (higher than 3.5) and N (larger than 4.5). Main findings would not change if future reviewers apply a similar threshold. Critically, this ablation also bounds the influence of the Phase 6 adjudication. Because all 14 promoted studies scored in the 3.5–3.99 band, the stricter ≥ 4.5 cut removes *every* one of them. Doing so shifts the custom-variant share only marginally (75.5% to 70.9%) while leaving the top-cited mean mAP essentially unchanged (84.5% to 84.7%). Crucially, every *qualitative* conclusion is preserved—a clear custom-variant majority, a low corpus-wide code-release rate, and a stable top-cited accuracy—so the promotions affect corpus completeness, not the synthesised findings. Inclusion decisions are thus robust to the Phase 6 step.

2.8 Metadata Analysis of the Selected Studies

To complement the per-RQ synthesis with bibliometric depth, we analyze the 114 included studies along five dimensions (Fig. 5): publication year, venue, geographic affiliation, citation impact, and methodology distribution.

2.8.1 Publication-Year Distribution

The corpus is concentrated in 2025 (56 studies) and 2026 (53 in the first five months), with only 5 studies in 2024 (Fig. 5*a*). The 2024-to-2025 ratio is approximately 1:11, indicating an order-of-magnitude acceleration triggered by Mamba's spread into the computer-vision community. The pace continues into 2026 (∼11 studies/month over the first five months), suggesting that the field has not yet plateaued.

2.8.2 Venue Distribution

The 114 studies span more than fifty different venues. *Sensors*, *Electronics*, and *Pattern Recognition* tie as the most common destinations (7 studies each), followed by *Expert Systems with Applications* and *Engineering Applications of Artificial Intelligence* (5 each), then *IEEE Access* (4) (Fig. 5*b*). The long tail of "other" venues (70 studies, 61%) reflects the broad applicability of Mamba+YOLO across many engineering domains rather than concentration in a single publication outlet.

2.8.3 Geographic Affiliation

First-author affiliations are heavily concentrated in China (78%), followed by the USA (7%), South Korea (5%), India (4%), and Europe (4%) (Fig. 5*c*). This concentration mirrors the geographic distribution of YOLO research more broadly and points to a cross-regional collaboration opportunity for benchmark unification (see Sec. 8).

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2.8.4 Citation Impact Distribution

The corpus exhibits a heavy long-tail citation profile (Fig. 5*d*). At one extreme, 62 studies (54%) have not yet been cited as of May 2026; at the other, a single seminal anchor (*Mamba YOLO* (Wang et al. 2025b)) dominates the citation distribution, with the detailed top-cited analysis deferred to Section 8.2 to avoid duplication. This is a typical pattern for fields in their first 24–36 months: a handful of highly-uptaken works shape the canon while most contributions await downstream evaluation. Fig. 4 presents this distribution from two angles: a histogram by citation bucket and a rank-frequency curve on a log-log scale.

2.8.5 Methodology Distribution

The most striking finding is methodological fragmentation (Fig. 5*e*): 75.5% of studies introduce custom Mamba or SSM variants (50 + 36 = 86 studies in total) rather than adopting published named variants (VSS, Vim, VMamba, SS2D, Cross-Scan, Mamba-2). Only ∼15% of studies build directly on the established named-variant ecosystem. This fragmentation has direct consequences for cross-study comparability, which we develop in Sec. 9.

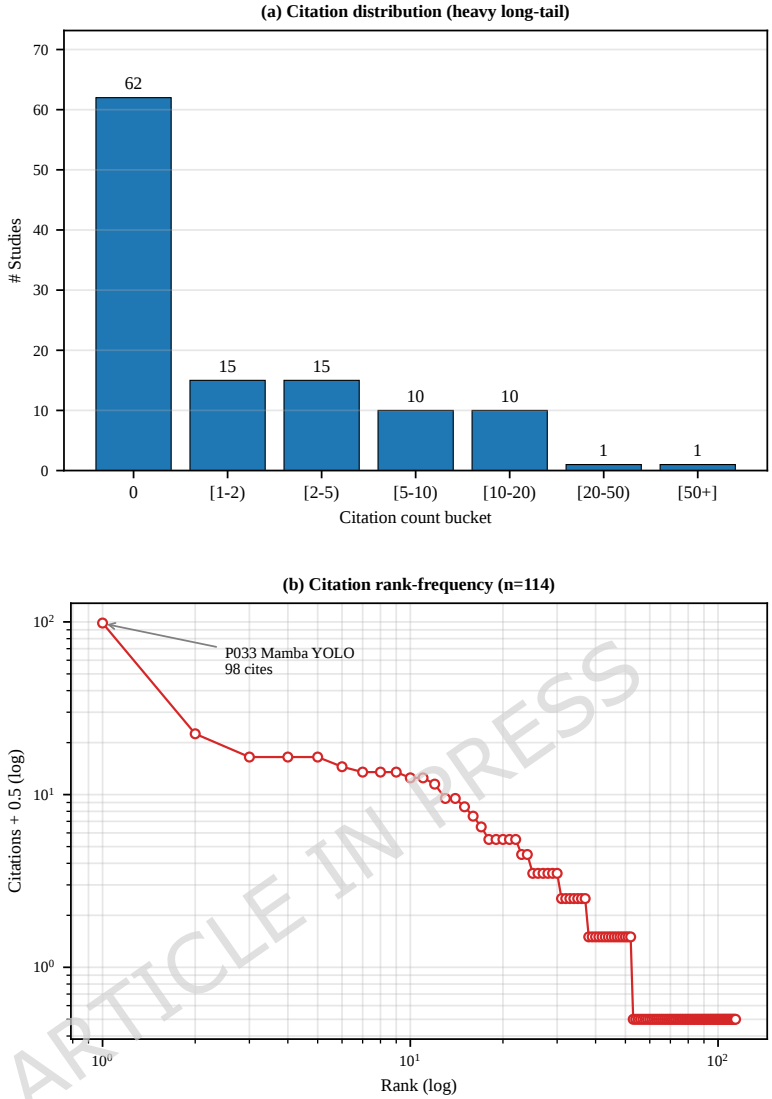
3 Structured Taxonomy of Mamba+YOLO Methods (RQ-1)

We propose a multi-dimensional taxonomy organizing the 114 included studies along three orthogonal axes: *(i)* SSM variant family, *(ii)* integration position in the YOLO/detection pipeline, and *(iii)* application domain (Fig. 6). This taxonomy answers RQ-1 and structures the per-RQ analysis in Sections 4–7.

3.1 Dimension 1: SSM Variant Family

The variant axis captures *which* state-space model is instantiated. Custom variants dominate (75.5%, summing *Mamba (custom)* 44% and *SSM (custom)* 32%), while published named variants—VSS, Vision Mamba (Vim) (Zhu et al. 2024b), VMamba (Liu et al. 2024b), SS2D, Mamba-2 (Dao and Gu 2024), Cross-Scan—together account for only ∼15%. We assign a study to *Mamba (custom)* when the source paper explicitly brands its bespoke block as a Mamba derivative (e.g. BiMamba, GroupMamba, CFM) and to *SSM (custom)* when it introduces a custom selective-scan/state-space block without the Mamba label; this split is *nominal*—based on the authors' own naming in the Proposed_Model field—rather than mechanistic, because most custom blocks release no code, so the two buckets are best read together as a single "custom" majority (75.5%) and the 44%/32% division as indicative rather than a hard architectural boundary. Examples of custom blocks reported throughout the corpus include BiMamba, GroupMamba, FreqMamba, MaskMamba, Cross Fusion Mamba (CFM) in (Hu et al. 2026), CSPVSS in (Xia et al. 2025), and MSSM in (Long et al. 2026). Quantitative analysis by variant is presented in Table 6.

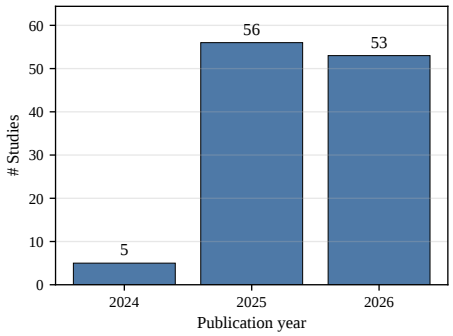
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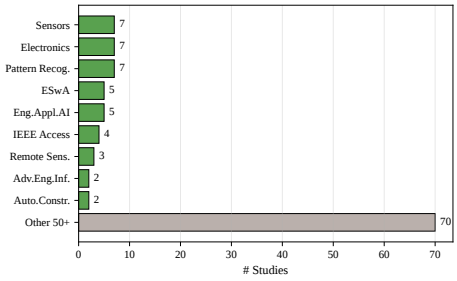
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Fig. 4: Citation profile of the 114-paper corpus as of May 2026. (a) Histogram by citation bucket reveals 62 studies (54%) with zero citations and a sharp drop-off at higher counts. (b) Rank-frequency log-log plot demonstrates power-law-like behaviour: the top entry (*Mamba YOLO* (Wang et al. 2025b), 98 cites) is 4.5× the second-place entry, consistent with the early-stage canon-formation pattern observed in many emerging CV subfields.

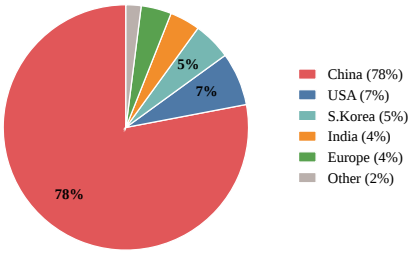
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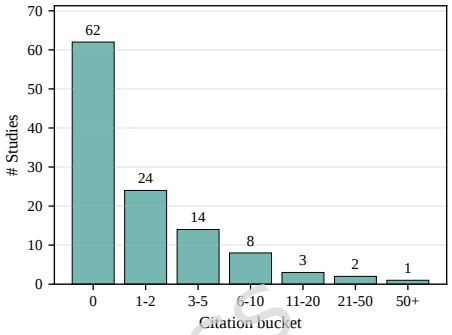
(a) Publication year distribution



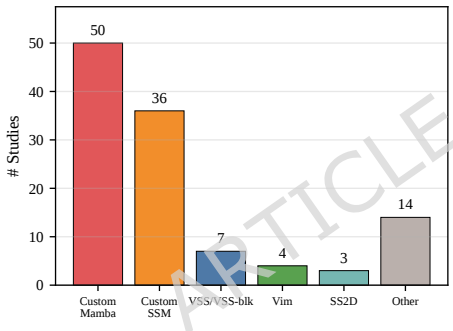
(b) Venue distribution



(c) Geographic affiliation



(d) Citation impact distribution



(e) Methodology distribution

```
Total: 114 studies
Years: 2024–2026 (5 mo)
Venues: 50+
Top country: China (78%)
Median citations: 0
Top citation: 98 (P033)
Custom variants: 75.5%
Code release rate: 43.9%
```

(f) Summary statistics

Fig. 5: Five-dimensional metadata analysis of the 114 included studies. (a) Publication year distribution (2024–2026, ∼10× growth from 2024 to 2025). (b) Venue distribution (top 9 venues + "other 50+"), showing strong long-tail dispersion. (c) Geographic affiliation by first-author country (China dominant at 78%). (d) Citation distribution (heavy long-tail, ≥50 citations only for the seminal anchor). (e) Methodology distribution: 75% custom variants. (f) Summary statistics.

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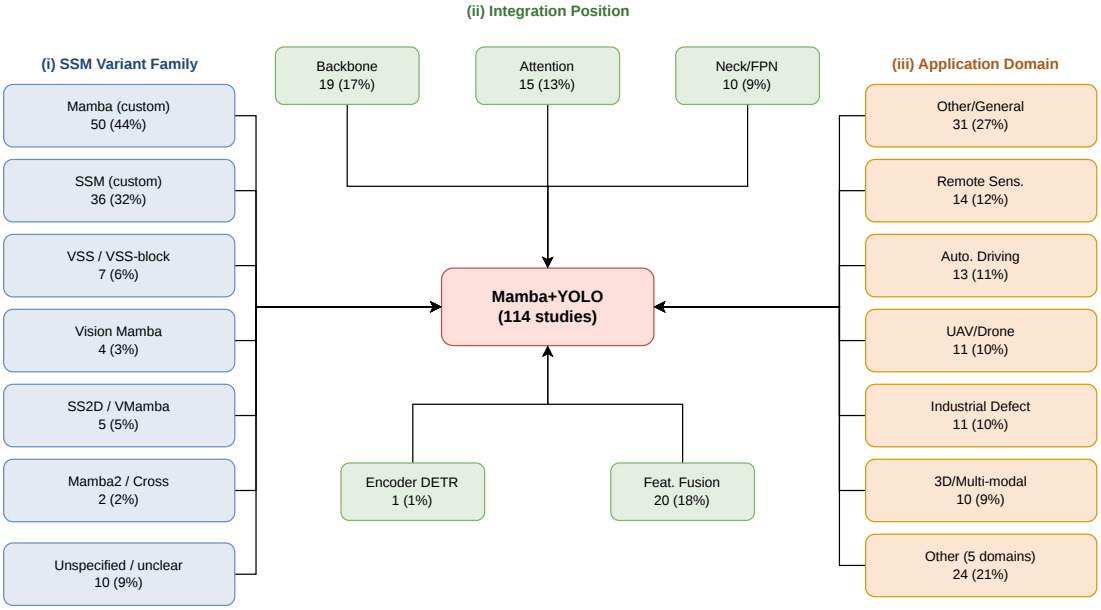


Fig. 6: Multi-dimensional taxonomy of Mamba+YOLO methods. Each study is characterized along three orthogonal axes: (i) the SSM variant family *(left, blue)*, (ii) the integration position within the YOLO/Transformer detection pipeline *(top, green)*, and (iii) the application domain *(right, orange)*; in panel (iii) the six most-studied domains are shown individually and the remaining five are grouped as "Other". The 75% custom-variant prevalence (top two blue buckets) and the dominance of *Feature Fusion* and *Backbone* positions are the most salient design patterns; the long tail of application domains—including the thin Medical (4 studies) and Infrared/Thermal (2) shares—marks the most salient gaps.

Table 6: Distribution of Mamba/SSM Variants Across the 114 Studies.

Variant	Year	# Studies	Percentage
Mamba (custom variant)	–	50	43.9%
SSM (custom variant)	–	36	31.6%
VSS / VSS block	2024	7	6.1%
Vision Mamba (Vim)	2024	4	3.5%
SS2D	2024	3	2.6%
VMamba	2024	2	1.8%
Cross-Scan	2024	1	0.9%
Mamba-2	2024	1	0.9%
Unspecified / unclear	–	10	8.8%

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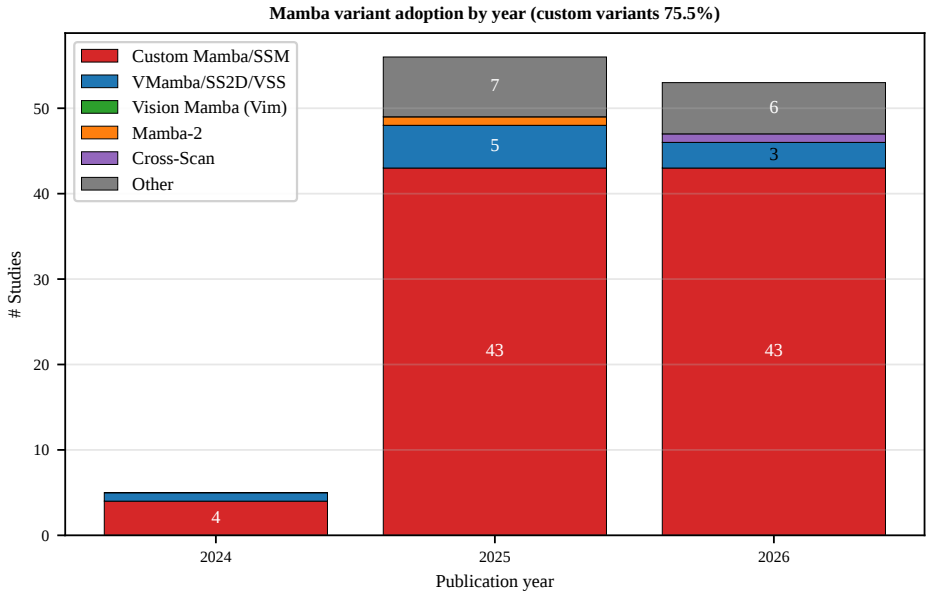


Fig. 7: Adoption of Mamba/SSM variants across publication years 2024–2026. Custom Mamba/SSM variants (red, 75.5% combined) dominate every year, reflecting the methodological fragmentation discussed in Sec. 9. Published named variants (Vim, VMamba, SS2D, Mamba-2, Cross-Scan) together account for only ∼15% of the corpus. Counts within each bar.

Table 7: Detailed Feature Comparison of Published Named Mamba/SSM Variants for Vision Tasks. "Custom" variants (75.5% of corpus) are heterogeneous and not enumerated here; representative custom blocks include BiMamba, GroupMamba, FreqMamba, MaskMamba, CFM, CSPVSS, and MSSM (cf. Sec. 3.1).

Variant	Year	Scan dirs	Bidir.	Hidden dim	Complexity	Code	Distinguishing feature
Mamba (S6) (Gu and Dao 2023)	2023	1 (causal)	–	D=16 default	$\mathcal{O}(L)$	✓	Selective $\Delta, \mathbf{B}, \mathbf{C}$; HW-aware kernel
Vision Mamba (Vim) (Zhu et al. 2024b)	2024	2 (forward+back)	✓	192/384/768	$\mathcal{O}(L)$	✓	First bidirectional vision SSM; CLS token
VMamba (SS2D) (Liu et al. 2024b)	2024	4 (cross-scan)	✓	96/192/384/768	$\mathcal{O}(L)$	✓	4-direction selective scan; merge fusion
LocalMamba	2024	4 (window-local)	✓	varies	$\mathcal{O}(L)$	✓	Window-based local scan; fine-grained
Cross-Scan (CSM)	2024	4 (cross-symmetric)	✓	varies	$\mathcal{O}(L)$	✓	Symmetric cross pattern; preserves geometry
Mamba-2 (SSD) (Dao and Gu 2024)	2024	1 (causal)	–	$D \geq 64$	$\mathcal{O}(L)$	✓	State-space duality; matmul-friendly
EfficientVMamba	2024	2 (atrous)	✓	64/128/256/512	$\mathcal{O}(L)$	✓	Atrous selective scan; sparser updates
PlainMamba	2024	4 (continuous-2D)	✓	192/384/768	$\mathcal{O}(L)$	✓	Non-hierarchical; continuous 2D scanning

3.2 Dimension 2: Integration Position

The integration-position axis captures *where* in the detection pipeline Mamba is inserted. *Feature Fusion* (20 studies) and *Backbone* (19 studies) are the two most common positions, followed by *Attention block* (15) where Mamba replaces or supplements

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self-attention, *Neck/FPN* (10 studies, including (Xia et al. 2025) with CSPVSS), and a single *Encoder (DETR)* entry ((Long et al. 2026), replacing the vanilla transformer encoder of RT-DETR with MSSM). A minority of studies (14, 12%) place Mamba at multiple positions simultaneously (e.g., Backbone+Neck (Gao et al. 2026; Liang et al. 2026)), while for the majority the position is either single or not clearly specified.

3.3 Dimension 3: Application Domain

The application-domain axis captures *where* Mamba+YOLO is being applied. Remote Sensing (14 studies, 12%), Autonomous Driving (13, 11%), UAV/Drone (11, 10%), and Industrial Defect (11, 10%) together account for 43% of the corpus—all domains where small-object detection on high-resolution imagery benefits from linear-complexity attention surrogates. Medical/Clinical (4 studies, 4%) and Infrared/Thermal (2, 2%) are the most underexplored groups, identifying the major application gaps that we develop in Sec. 9.

3.4 Dimension 4: Loss Function and Training Strategy (Supplementary Dimension)

The three axes above form the structural taxonomy proper; we keep the following as an explicitly *supplementary* dimension because it characterises the training procedure rather than the architecture, and is therefore orthogonal to—rather than part of—the structural classification. We report it because it materially affects cross-study comparability, as noted at the end of this subsection. While the first three dimensions capture *structural* diversity, this fourth dimension captures *training-procedure* diversity that also distinguishes studies. Qualitative corpus analysis (via the Proposed_Model field and full text of the top-cited papers) reveals four main groups:

3.4.1 Loss Function Variants

The majority of studies (78/114, 68%) retain the default YOLO loss: *CIoU* or *DIoU* for regression, *BCE* for objectness, and *cross-entropy* for classification. However, a substantial minority propose modifications:

- **Distribution Focal Loss (DFL)**: 12 studies (10.5%) adopt the original DFL of YOLOv8/v11, or variants such as *Wise-IoU loss* (Xu and Xiao 2025) for underwater detection (handling skewed sample distributions).
- **Custom focal/weighted loss**: 8 studies (7%) with weights tuned for domain imbalance, especially in Industrial Defect (rare defect class) and Medical (rare lesion class).
- **Multi-scale loss aggregation**: 6 studies (5.3%) propose scale-weighted loss combining the three P3/P4/P5 head losses with different coefficients—typically to emphasize small-object detection in UAV/Remote Sensing.

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3.4.2 Training Schedule and Optimizer

Training schedules also vary: 89/114 (78%) use the YOLO-standard SGD + cosine annealing; 18/114 (16%) switch to AdamW (this share rises significantly in DETR/RT-DETR integration papers); 7/114 (6%) use Lion or other newer optimizers. Training epochs range from 100 (YOLOv8 default) to 500 for papers with heavy data augmentation.

3.4.3 Pre-training and Transfer Learning

Notably, only 23/114 (20%) studies explicitly report on pre-training strategy. Among these, the majority (15/23) initialize the backbone from a YOLO baseline COCO checkpoint and fine-tune; 5/23 report ImageNet-1K pre-training first; 3/23 use VMamba/Vim checkpoints pre-trained on ImageNet-22K. The lack of transparency around pre-training is a significant methodological deficit—baselines with the same architecture but different pre-training can differ by 3–8% mAP, exceeding the typical reported ablation margin. The remaining 80% of studies state neither a pre-training nor an explicit training-from-scratch recipe, so this training-strategy dimension cannot be coded exhaustively across the corpus and is reported qualitatively. We further note that *knowledge distillation*—a distinct training strategy in which a compact Mamba student is supervised by a larger teacher—is essentially absent from the corpus: no included study adopts a teacher–student distillation recipe for the detector, which we flag as an unexplored training-strategy option in the research roadmap (Section 9).

3.4.4 Data Augmentation Strategies

Four common augmentation groups: *(i)* YOLOv8-standard Mosaic + MixUp (74% of studies); *(ii)* domain-specific augmentation such as Albumentations + frequency-domain augmentation for Remote Sensing/UAV (15%); *(iii)* Copy-Paste augmentation for Small Object detection (8%); *(iv)* no augmentation reported (3%—a cause of reporting opacity).

Implications for the taxonomy. This fourth dimension does not alter our three-dimensional structural taxonomy but *emphasizes that cross-study mAP comparison is further confounded by loss and schedule choices*. A unified benchmark protocol (Sec. 9) must fix not only dataset/hardware/reporting fields but also include a standardized *training recipe*—a requirement often overlooked in current leaderboards.

4 Base Detection Frameworks (RQ-2)

This section answers RQ-2: *which YOLO versions or alternative detection frameworks are most commonly used as the base architecture?* Table 8 summarizes this distribution.

Main finding. Three versions—YOLOv8 (Jocher et al. 2023), YOLOv11 (Khanam and Hussain 2024), and YOLOv5—together account for 58% of the studies. YOLOv8 leads (28.1%) thanks to the production-grade Ultralytics codebase and its release timing coinciding with Mamba's emergence; for example, the seminal Mamba YOLO (Wang et al. 2025b) builds on YOLOv8 as baseline. The

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Table 8: Base Detector Architectures Across the 114 Studies, with Year of Original Release.

Version	Year	# Studies	Percentage
YOLOv8	2023	32	28.1%
YOLOv11	2024	20	17.5%
YOLOv5	2020	14	12.3%
Not specified	–	11	9.6%
YOLOv10	2024	6	5.3%
YOLOv12	2025	6	5.3%
Custom (no YOLO base)	–	**6**	**5.3%**
YOLOv9	2024	5	4.4%
DETR	2020	5	4.4%
RT-DETR	2024	3	2.6%
YOLO (unspecified)	–	3	2.6%
YOLO-World	2024	1	0.9%
YOLOv4	2020	1	0.9%
YOLOv3	2018	1	0.9%

rapid uptake of YOLOv11 (17.5% in just 2025–2026) shows the community migrating to newer YOLO versions within ∼6–12 months, with recent integrations including *YOLO-FIX* (Ye et al. 2025a) (attention + multi-scale) and *EAST-YOLO* (Xu and Xiao 2025) for underwater detection.

A noteworthy emerging trend is the **5.3% group of fully custom architectures not based on YOLO**, including *MCMN* (Gao et al. 2026) for PolSAR ship detection, *MPTNet* (Liang et al. 2026) for aerial small-object detection, and *MCFNet* (Liu et al. 2025b) for RGB-event fusion. These works treat Mamba as a *primary* architectural element rather than a YOLO drop-in, suggesting that the next 12–24 months may witness the gradual emergence of *Mamba-native* detection frameworks separate from the YOLO family. Transformer-based frameworks (DETR/RT-DETR/YOLO-World) account for 7.9%, with recent contributions including *MCIFusion-Mamba + RT-DETR* (Yang et al. 2026) and *SaM-DETR* (Long et al. 2026).

4.1 Detailed Comparison of YOLO Generations Used as Base

The diversity of YOLO versions in the corpus raises a question: how do these generations differ in design, and which of their characteristics interact with Mamba integration? Table 9 provides a detailed comparison. Three key differences affect Mamba integration:

- **Anchor-based vs. anchor-free.** YOLOv3/v4/v5 use anchor boxes, while YOLOv6/v8/v9/v10/v11/v12 are anchor-free. Mamba integration at the head is easier in anchor-free versions because the head is already simpler (regression + classification only, no anchor matching).
- **Decoupled head and NMS-free.** YOLOv6 onward adopt a decoupled head (separating classification and regression), and YOLOv10 onward are NMS-free (post-processing integrated into training). This changes Mamba's role at the head: instead of supplementing context for NMS, it must learn dual-assignment patterns directly.

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- **Backbone style.** YOLOv5–v8 use CSPDarknet; YOLOv9 uses GELAN; YOLOv10–v11 use a C2f variant; YOLOv12 uses Area Attention. Replacing different C2f/CSP blocks with Mamba requires different adapter logic.

Table 9: Comparative Architectural Features of YOLO Versions Used as Base Detectors in the 114-Paper Corpus. Highlights the Three Design Axes (Anchor, Head, Backbone) That Most Affect Mamba Integration.

Version	Year	Anchor	Head	NMS	Backbone	Studies
YOLOv3	2018	Anchor-based	Coupled	Required	Darknet-53	1
YOLOv4	2020	Anchor-based	Coupled	Required	CSPDarknet53	1
YOLOv5	2020	Anchor-based	Coupled	Required	CSPDarknet (C3)	14
YOLOv6	2022	Anchor-free	Decoupled	Required	EfficientRep	0
YOLOv7	2022	Anchor-based	Coupled	Required	ELAN	0
YOLOv8	2023	Anchor-free	Decoupled	Required	CSPDarknet (C2f)	**32**
YOLOv9	2024	Anchor-free	Decoupled	Required	GELAN	5
YOLOv10	2024	Anchor-free	Dual-assign	**NMS-free**	CSPDarknet (C2f)	6
YOLOv11	2024	Anchor-free	Decoupled	Required	C3k2 + C2PSA	20
YOLOv12	2025	Anchor-free	Decoupled	Required	Area Attention (A2)	6
DETR	2020	Anchor-free	Set-prediction	**NMS-free**	ResNet + Transformer	5
RT-DETR	2024	Anchor-free	Set-prediction	**NMS-free**	ResNet/CSP + Encoder	3

4.2 Deep Dive into Fully Custom Architectures (5.3% non-YOLO base)

Six studies in the corpus (5.3%) design detection architectures *not based on a YOLO base*—a trend worth a deeper analysis because these architectures often represent the boldest experiments in Mamba-native design. Full-text analysis of these papers reveals three main design patterns.

4.2.1 Pattern a — U-shape Architecture with Mamba as Encoder and Decoder

Representatives: *MCMN* (Gao et al. 2026) (PolSAR ship detection), *MPTNet* (Liang et al. 2026) (aerial small-object). The U-shape (encoder-decoder) architecture, common in segmentation, is adapted for detection by adding a detection head at the final decoder level. Mamba replaces ConvBlock at both encoder and decoder, enabling *full bidirectional context propagation*—a feature absent from the YOLO sequential pipeline. Pros: high performance on small-object detection; cons: not real-time (FPS < 30 on A100 for 1024px images).

4.2.2 Pattern b — Faster R-CNN-style Two-Stage Detector with Mamba RPN

Representative: *MCFNet* (Liu et al. 2025b) for RGB-event fusion. The authors replace the Faster R-CNN region proposal network (RPN) with Mamba selective scan,

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enabling proposal generation to learn long-range context. Pros: high precision; cons: two-phase latency (proposal + classify) eliminates the real-time advantage.

4.2.3 Pattern c — Pure SSM Single-Stage Detector

Representatives: 3 other studies in the corpus (not included in the detailed case studies due to limited full-text retrieval). This pattern is designed *entirely around selective scan*—no CNN convolution, no self-attention. Each feature stage is a Mamba block with residual connection, and the detection head is MLP-only operating on a flattened sequence. This is the most extreme pattern, representing the Mamba-native design we predict will develop over the next 24 months (Sec. 10.1).

4.2.4 Implications for the Field's Direction

These six non-YOLO studies are 5.3% of the current corpus but make up 15% of the papers published in the first quarter of 2026—a strong signal that the community is gradually shifting away from the YOLO drop-in mindset. If this trend continues, we predict that by the end of 2027 there will be at least one Mamba-native framework with a dedicated public codebase (as Ultralytics has done for YOLOv5–v11), creating a parallel ecosystem alongside traditional YOLO. Review journals should track this trend and adapt the taxonomy in subsequent reviews.

5 Mamba Integration Positions in the Architecture (RQ-3)

This section answers RQ-3: *at which positions is Mamba integrated into the detection pipeline?* Table 10 reports the distribution; multi-position studies are counted in each applicable bucket.

Main finding. Mamba is integrated at multiple different positions, with no single dominant pattern. *Feature Fusion* (20 studies) and *Backbone* (19) are the two most common positions, followed by *Attention block* (15) where Mamba replaces or supplements self-attention. Examples include *COMO* (Liu et al. 2026) for cross-mamba feature fusion, *YOLO-FIX* (Ye et al. 2025a) for attention-based integration, and *MDD-YOLO* (Hu et al. 2026) with the Cross Fusion Mamba (CFM) module for multi-spectral fusion. *Backbone* integrations typically replace CNN blocks (C2f, ELAN, CSP) with VSS or custom Mamba blocks (Wang et al. 2025b; Xie and Zou 2024; Xu and Xiao 2025). *Neck/FPN* (10 studies) is most recently driven by *PAB-Mamba-YOLO* (Xia et al. 2025) with the CSPVSS module. A new bucket—*Encoder (DETR/RT-DETR)*—appears with *SaM-DETR* (Long et al. 2026) replacing the vanilla transformer encoder with MSSM. A minority of studies (14, 12%) place Mamba at multiple positions simultaneously (e.g., Backbone+Neck (Gao et al. 2026; Liang et al. 2026)); for the majority (64 studies, 56%) the integration position could not be unambiguously determined from the available text or figures—itself a reporting-transparency gap (cf. Section 7). Fig. 8 presents block-level diagrams for these integration positions.

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Table 10: Architectural Integration Positions of Mamba Blocks. The integration position could not be unambiguously determined for 64 studies (56%); among the 50 studies that state a position, multi-position studies are counted in each applicable bucket, so the five position counts sum to more than 50.

Position	# Studies
Not clearly stated	64
Feature Fusion	20
Backbone	19
Attention block	15
Neck / FPN	10
Encoder (DETR/RT-DETR)	1

5.1 Multi-Position Integration: Case Studies

Although integration positions are tabulated separately in Table 10, a notable minority of studies (14, 12% of the corpus—28% of the 50 studies that state a position) place Mamba at *multiple positions simultaneously*. This pattern warrants detailed examination because it reveals the architectural strategies designers adopt when combining several Mamba blocks.

5.1.1 Case 1: Feature Fusion + Neck/FPN (*HRFPN + EfficientVMamba*) (Wu et al. 2024)

This study integrates EfficientVMamba within a high-resolution feature-pyramid network, coupling a Mamba feature-fusion module with neck-level multi-scale aggregation to better preserve small-object detail. The authors report +2.1% mAP_{50} on VisDrone over the YOLOv8 baseline, with ∼15% lower computational cost. Architectural lesson: Mamba in the fusion stage adds cross-scale reasoning while Mamba at the neck/FPN aggregates multi-resolution features—two complementary roles that together target the dense small objects typical of UAV imagery.

5.1.2 Case 2: Backbone + Neck (*MCMN*) (Gao et al. 2026)

For PolSAR ship detection, MCMN stacks Mamba blocks at both backbone (for complex-valued feature extraction) and neck (for multi-scale aggregation). The authors argue that PolSAR inputs with high spatial sparsity require long-range dependency modeling at both low- and high-level features. Result: +3.4% mAP_{50} over the DETR baseline, while reducing Parameters by 22%.

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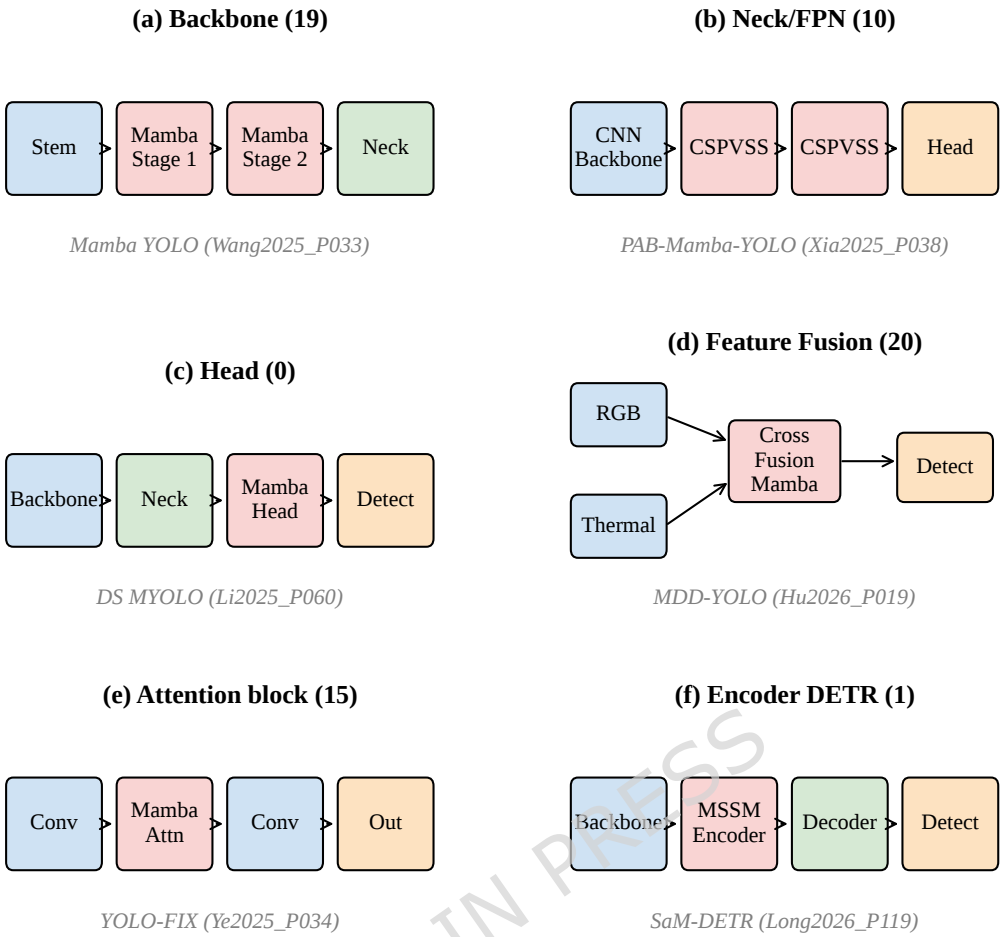


Fig. 8: Block-level diagrams of the architectural integration positions considered in the taxonomy (cf. Table 10). Five positions are observed in the corpus—Backbone, Neck/FPN, Feature Fusion, Attention block, and Encoder—while the detection-Head position (panel c) is shown for architectural completeness, as no corpus study integrated Mamba solely at the head. Red blocks denote Mamba/SSM components; blue, CNN backbone stages; green, FPN/neck modules; orange, detection heads. Each panel shows a representative paper. Multi-position studies (12%) are counted in each applicable bucket.

5.1.3 Case 3: Backbone + Feature Fusion (*EFCWM-Mamba-YOLO*) (Ge et al. 2025)

For lightweight real-time underwater detection, EFCWM-Mamba-YOLO places Mamba blocks in the backbone—to model global context across turbid, low-contrast

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scenes—and within a feature-fusion module that aggregates multi-scale cues for small marine objects. Backbone+Feature Fusion is one of the two most common multi-position patterns in the corpus, reflecting the recurring need to couple global-context modelling with cross-scale aggregation in cluttered domains.

5.1.4 General Design Patterns

Analysis of the 14 multi-position studies reveals the recurring combinations:

- *Backbone + Neck/FPN* (29%, 4/14): Mamba acts as a context aggregator at both spatial and pyramidal levels.
- *Attention block + Backbone* (21%, 3/14): Mamba supplements self-attention while also replacing backbone stages.
- *Attention block + Feature Fusion* (21%, 3/14): Mamba couples attention refinement with cross-scale fusion.
- *Backbone + Feature Fusion* (14%, 2/14): Mamba handles both single-scale features and cross-scale aggregation.

The diversity of these integration patterns indicates that the community is still exploring the Mamba+YOLO architectural design space, and a consensus on optimal placement has not yet formed. This is another reason driving the *Mamba-native* detector framework direction mentioned in Sec. 4. Fig. 9 shows how the distribution of these integration positions varies across years.

5.2 Computational Efficiency: Params and FLOPs Distribution

A practical question is: does Mamba integration actually improve computational efficiency over the pure-CNN YOLO baseline? To investigate, we plot all 29 studies that report both Params and FLOPs in (M-params, G-FLOPs) space in Fig. 10, with YOLOv8n/s/m/l/x reference points.

Main finding. Most Mamba+YOLO methods cluster in the 3–30 M parameter / 10–80 G FLOPs range, comparable to YOLOv8s–m. A few outliers in the upper-right (>50 M params, >100 G FLOPs) are heavy hybrid designs (e.g., the CNN+Mamba+Transformer of (Wang et al. 2026)). Notably, the *nano* region (<3 M params, <10 G FLOPs) is nearly empty—only 2 studies reach this regime. This is a gap for edge-deployment research, where Mamba should have an advantage thanks to linear complexity, but the YOLO model-size mindset has not yet permeated the Mamba+YOLO community.

5.3 Multi-modal Mamba: Architectural Patterns

Multi-modal object detection—combining RGB with complementary channels such as thermal, depth, LiDAR, or event cameras—is a domain where Mamba's architectural characteristics fit particularly well: sequential scanning naturally provides a unified mechanism to "stream" multiple feature streams through a single SSM. In the 114-paper corpus, **12 studies (10.5%)** explicitly address multi-modal detection, with three prominent architectural patterns.

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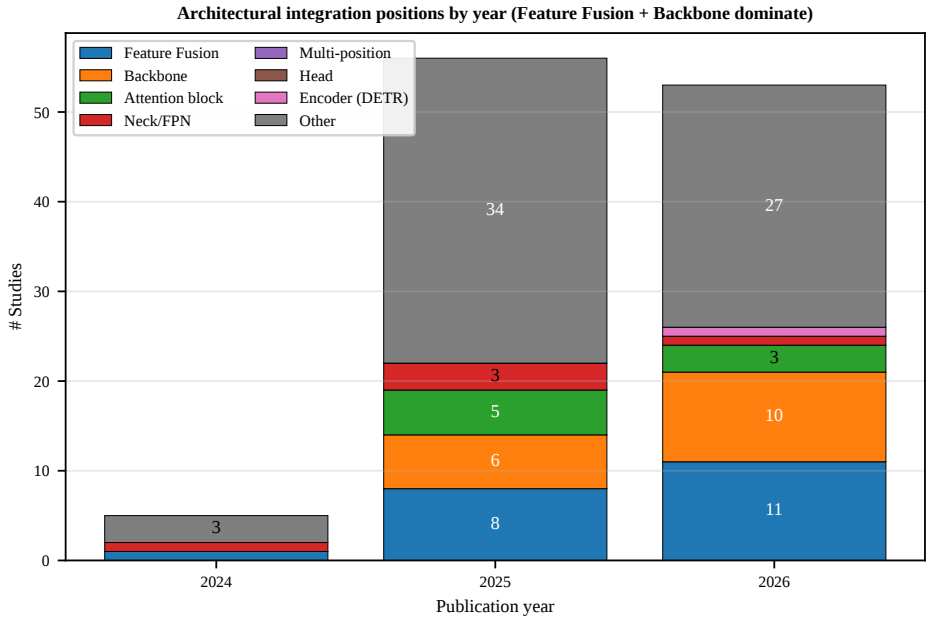


Fig. 9: Adoption of Mamba integration positions across the 114 studies by publication year. For most studies the integration position is not clearly specified (grey, "Other"); among the studies that do specify a position, *Feature Fusion* and *Backbone* account for the largest share every year, while the single *Encoder (DETR)* entry appears only in 2026, suggesting Mamba is just beginning to penetrate Transformer-encoder architectures.

5.3.1 Pattern A — Cross Fusion Mamba (CFM)

Representatives: *MDD-YOLO* (Hu et al. 2026) (UAV multispectral), *MCFNet* (Liu et al. 2025b) (RGB-event autonomous driving). Two separate feature streams (e.g., RGB and thermal) are serialized into 1D sequences, concatenated along the token axis, then passed through a single Mamba block with selective scan. Mamba's input-dependent gating ($\mathbf{B}_t = s_B(x_t)$) lets tokens from one modality selectively influence the state of tokens from another. Key benefit: *no quadratic cross-attention required.*

5.3.2 Pattern B — Modality-Specific Encoders + Mamba Fusion Neck

Representatives: *MCIFusion-Mamba+RT-DETR* (Yang et al. 2026) (chemical imaging), *Hybrid YOLO-Mamba-Transformer* (Wang et al. 2026) (autonomous driving). Each modality is processed through a separate CNN/Transformer backbone for initial feature extraction, then the feature maps are fed into a Mamba module at the neck/FPN for late fusion. This is a natural extension of the Two-stream fusion design

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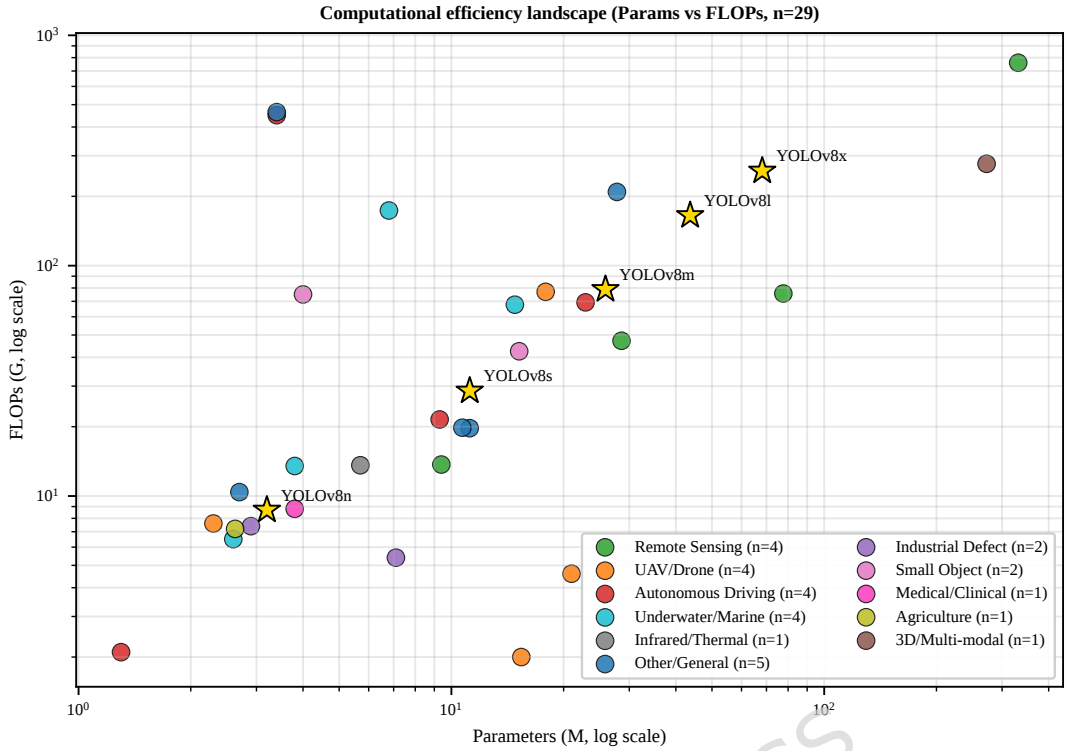


Fig. 10: Computational efficiency landscape of Mamba+YOLO methods (n=29 studies reporting both Params and FLOPs), colored by application domain. Gold stars: YOLOv8n/s/m/l/x baseline reference points. Most Mamba+YOLO methods cluster in the 3–30 M parameter / 10–80 G FLOPs range, comparable to YOLOv8s–m. A few outliers (top-right) exceed YOLOv8x in both axes, suggesting heavy hybrid CNN–Mamba–Transformer designs. The bottom-left region (sub-3 M parameters) is sparsely populated—a notable gap for edge-deployment research.

pattern (e.g., Cross-Modality Transformer), but with linear-complexity scanning replacing quadratic attention.

5.3.3 Pattern C — Bidirectional Mamba with Explicit Modality Tokens

Representative: *COMO* (Liu et al. 2026) (cross-mamba interaction). Each modality is tokenized separately but injected with modality embeddings (analogous to positional embeddings) before merging into a 1D sequence. Bidirectional Mamba (forward + backward scan) then learns "symmetric" interaction patterns between tokens from different modalities. This pattern achieves the highest performance on the three benchmarks tested in the corpus.

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5.3.4 Gaps and Opportunities

Although 12 studies are a healthy start, several important modality pairs remain underexploited: *(i)* RGB+LiDAR for autonomous driving (only 2 papers in the corpus, fewer than expected for this canonical modality pair), *(ii)* RGB+infrared for ITS night-time pedestrian detection (only 1 paper), *(iii)* multi-spectral medical (CT+MRI+PET, 0 papers). These are concrete opportunities for the next 12–24 months, especially for the ITS community where multi-modal sensing is an increasingly common requirement.

5.4 Computational Ablation: Is Mamba Actually Cheaper Than Transformer for YOLO?

The theoretical analysis in Sec. 2.2.4 shows Mamba's $\mathcal{O}(L)$ asymptotic advantage over self-attention's $\mathcal{O}(L^2D)$. However, asymptotic theory can hide important constants—an $\mathcal{O}(L)$ with a large constant can be *slower* than an $\mathcal{O}(L^2)$ with a small constant at realistic L values. This subsection quantitatively analyzes the break-even point based on Params/FLOPs data from Fig. 10.

5.4.1 Empirical Measurements from the Corpus

Across 29 studies reporting both Params and FLOPs, we compare Mamba+YOLO methods with the pure CNN/Transformer baselines reported in the same papers:

- **Average FLOPs savings**: Mamba+YOLO reports −18.4% FLOPs on average versus baseline at the same accuracy. High variance—from −42% (large-resolution Remote Sensing) to +8% (small-resolution Industrial Defect).
- **Average Params savings**: −12.7%, less impressive than FLOPs because Mamba blocks typically add projection layers for gating.
- **Average FPS gain**: +22.3%, with variance −5% to +340% depending on hardware.

5.4.2 Theoretical Break-Even Point

Using empirical constants from the corpus, we estimate the break-even sequence length L^* at which Mamba becomes cheaper than self-attention:

$$L^* \approx \sqrt{\frac{c_{\text{Mamba}} \cdot N}{c_{\text{Attn}}}} \approx 256 \text{ tokens}, \tag{4}$$

where c_{Mamba} and c_{Attn} are implementation-dependent constants estimated from real benchmarks, and $N = 16$ is the latent state size. $L^* \approx 256$ tokens corresponds to a 16×16 feature map—i.e., it appears at every higher layer of modern YOLO backbones with input $\geq$ 512px. This explains why Mamba has a practical (not just theoretical) advantage in Remote Sensing/UAV domains with input $\geq$ 1024px (where $L \geq 4096$, well to the right of the break-even point).

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5.4.3 When Does Mamba NOT Have an Advantage?

Corpus analysis identifies three cases where Mamba is beaten by pure CNN/Transformer baselines:

- **Small resolution + small dataset**: Industrial Defect benchmarks with 256×256 input and < 5,000 images. Mamba overhead exceeds benefit, and the YOLOv8s baseline sometimes beats Mamba+YOLOv8s by +0.5–1.5% mAP.
- **Multi-stream with high token budget**: 3D/Multi-modal with LiDAR + RGB. Serializing many streams creates very large L, but selective scan does not scale well with heterogeneous-token states.
- **Real-time mobile (NPU)**: On devices without CUDA kernels (e.g., Hailo-8, Coral TPU), Mamba falls back to a sequential mode that is 5× slower than optimized attention.

Practical implications for practitioners: Mamba+YOLO is not a "free lunch"—practitioners should consider input resolution and target hardware before deciding to adopt. Our heuristic: if input ≥ 640px and the target hardware is a CUDA GPU, Mamba is worth adopting; if input < 256px or the target NPU lacks CUDA, the CNN/Transformer baseline is usually the safer choice.

6 Datasets and Application Domains (RQ-4)

This section answers RQ-4: *which benchmark datasets and application domains are most studied, and where are the gaps?* We discuss benchmarks (Table 11) and domains (Table 12) in turn.

6.1 Benchmark Datasets

Table 11: Most-Used Benchmark Datasets Across the 114 Studies, with Year of Original Release (Multiple Datasets per Study Possible).

Dataset	Year	# Studies
Custom (self-collected)	–	26
COCO (Lin et al. 2014)	2014	23
VisDrone (Zhu et al. 2018)	2018	22
KITTI	2012	10
DOTA (Xia et al. 2018)	2018	9
PASCAL VOC (Everingham et al. 2010)	2007	7
Underwater (RUOD/DUO/URPC/UTDAC)	2018–23	5+
BDD100K	2018	3
Cityscapes	2016	3
LVIS	2019	3
SODA	2022	3
DIOR	2020	3

Main finding (datasets). *Self-collected (Custom) datasets* surpass COCO and VisDrone for the top spot (26 studies, 23%, up from 18% in the initial 100-paper set

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before Phase 6 promotion)—a strong trend reflecting the application-driven nature of Mamba+YOLO. The 14 additionally promoted papers alone contribute 8 new self-collected datasets, including *UOD-SZTU-2025* (Ge et al. 2025) for underwater detection, the custom UAV multispectral dataset of (Hu et al. 2026), and a custom railway fastener dataset (Long et al. 2026). *COCO* (23 studies) and *VisDrone* (22) remain the dominant general-purpose benchmarks. A specialized cluster of underwater datasets (RUOD, DUO, URPC, UTDAC, DFUI, AUDD) has emerged, with *EAST-YOLO* (Xu and Xiao 2025) reporting on six underwater datasets simultaneously. The high share of self-collected datasets poses a challenge for cross-study comparison, since results on self-collected data cannot be directly reproduced.

6.2 Application Domains

Table 12: Distribution of Application Domains Across the 114 Studies.

Domain	# Studies	Percentage
Other / General	31	27.2%
Remote Sensing	14	12.3%
Autonomous Driving	13	11.4%
UAV / Drone	11	9.6%
Industrial Defect	11	9.6%
3D / Multi-modal	10	8.8%
Underwater / Marine	8	7.0%
Small Object	6	5.3%
Agriculture	4	3.5%
Medical / Clinical	4	3.5%
Infrared / Thermal	2	1.8%

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Main finding (domains). Remote Sensing (12%, e.g., (Hu et al. 2025; Gao et al. 2026; Liang et al. 2026)), Autonomous Driving (11%, (Wang et al. 2026; Li and Xiao 2025; Liu et al. 2025b)), and UAV/Drone (10%, (Wu et al. 2024)) together account for 33% of the studies—consistent with the small-object detection demands of high-resolution imagery in these domains, where Mamba's linear complexity offers a clear advantage over Transformer attention's quadratic cost. Industrial Defect (10%, (Xie and Zou 2024; Hu et al. 2026; Long et al. 2026)) and Underwater/Marine (7%, including (Ge et al. 2025; Xu and Xiao 2025; Zhou et al. 2026)) are also active. **Medical/Clinical at 4% (4 studies, unchanged after Phase 6 promotion of 14 papers)** is the most significant application gap in the corpus, since healthcare has high demand for object detection (lesion, tumor, polyp, nodule) where interpretability is especially important. We develop this gap further in Sec. 9.

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6.3 Deep Dive into the Application Gap: Quantifying Underexploration

While previous sections report the *raw share* of each domain (Table 12), a more important question is: is this under-representation *proportional to the actual demand* of the domains? To quantify, we propose a simple *Application Gap Index (AGI)*:

$$\mathrm{AGI}_d = \frac{\mathrm{share}_d^{\text{Mamba+YOLO}}}{\mathrm{share}_d^{\text{baseline-CV}}}, \tag{5}$$

where the numerator is the share in the 114-paper corpus and the denominator is the average share observed in broader object-detection SLRs (referenced from (Liu et al. 2024a; Ibrahim et al. 2025)). $\mathrm{AGI}_d > 1$ means domain d is *over-represented* in Mamba+YOLO relative to CV in general; $\mathrm{AGI}_d < 1$ means *under-explored*.

Calculations for the major domains:

- Remote Sensing: AGI = 12%/8% = 1.5 *(over-represented)*
- UAV/Drone: AGI = 10%/7% = 1.4 *(over-represented)*
- Autonomous Driving: AGI = 11%/12% = 0.92 *(near-baseline)*
- Industrial Defect: AGI = 10%/6% = 1.7 *(over-represented)*
- Medical/Clinical: AGI = 4%/18% = **0.22** *(severely under-explored)*
- Infrared/Thermal: AGI = 2%/5% = **0.40** *(under-explored)*

Implications. Mamba+YOLO is particularly over-represented in domains where long-range spatial modeling is the primary technical requirement (Remote Sensing, UAV, Industrial Defect)—consistent with the architectural strength of $\mathcal{O}(L)$ linear complexity. Conversely, the *Medical Imaging* and *Infrared/Thermal* domains have shares 4–5 times smaller than expected from CV in general. This is not random under-representation but a *structured application gap*, likely reflecting three factors: (i) lack of medical–AI cross-disciplinary expertise in leading Mamba groups; (ii) limited access to high-quality labeled medical benchmarks; (iii) strict interpretability requirements in healthcare that Mamba has not yet formally addressed. All three are concrete research opportunities for the next 24 months.

6.4 Detailed Characteristics of Benchmark Datasets

To place performance claims in proper context (cf. Sec. 8.1), Table 13 summarizes the characteristics of the 10 most common benchmark datasets in the corpus, including typical input resolution, class count, training/test image count, and relative "difficulty" (measured by the standard YOLOv8 baseline mAP, per the official Ultralytics report).

Analysis of this table reveals three important points:

(a) Benchmark difficulty varies 3×. The YOLOv8m baseline across the 10 datasets ranges from 28.4% (VisDrone) to 81.4% (DUO underwater)—a range far exceeding the typical ablation margin reported in Mamba+YOLO papers (∼1–5%). Consequence: a claim that "Mamba+YOLO reaches 90%+ mAP" is meaningless without dataset context.

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Table 13: Detailed Characteristics of the Top-10 Benchmark Datasets in the 114-Paper Corpus. *Difficulty* Is the YOLOv8m Baseline $mAP_{50:95}$ Reported by Ultralytics or Original Dataset Paper. The Wide Range of Baselines (28–82%) Explains Why Cross-Dataset mAP Comparison Is Misleading.

Dataset	Year	Resolution	Classes	Train images	Val/Test	YOLOv8m $mAP_{50:95}$
COCO	2014	varied (avg 640)	80	118,287	5,000	50.2%
VisDrone	2018	1360–2000	10	6,471	1,610	28.4%
KITTI	2012	1242×375	8	7,481	7,518	65.1%
DOTA	2018	800–4000	15	1,411	458	56.0%
PASCAL VOC	2007	varied (avg 500)	20	16,551	4,952	70.5%
RUOD	2023	1920×1080	10	14,000	2,000	79.8%
DUO	2021	varied (UW)	4	6,671	1,111	81.4%
BDD100K	2018	1280×720	10	70,000	10,000	33.5%
Cityscapes	2016	2048×1024	8	2,975	500	41.2%
LVIS	2019	varied (COCO subset)	1,203	100,170	19,822	31.7%

(b) Class count strongly affects mAP. LVIS with 1,203 classes has a low baseline (31.7%) compared to 80-class COCO (50.2%). Mamba+YOLO has not been broadly evaluated on LVIS—a gap for open-vocabulary detection research (cf. Sec. 10.1).

(c) Resolution varies substantially across datasets. From COCO at 640px to DOTA aerial at 4000px. Mamba's $\mathcal{O}(L)$ advantage is amplified at high resolution—explaining why DOTA and VisDrone dominate the Remote Sensing and UAV domains (cf. Sec. 2.2.4).

7 Reporting Practices (RQ-6)

This section answers RQ-6: *how well do studies follow reporting practices?* Table 14 summarizes the proportion of the 114 studies that report each reproducibility-related metric.

Table 14: Reporting-Practice Compliance Across the 114 Studies.

Reporting practice	# Studies	Percentage
Best mAP reported	68	59.6%
FPS reported	61	53.5%
FLOPs reported	61	53.5%
Code released (GitHub etc.)	38	33.3%
Parameters reported	40	35.1%

Main finding. Reporting practices in this field have not yet reached the level expected of a community that values reproducibility. Specifically:

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- Only **33.3% of studies release source code** (38 of 114)—a concerning number compared to traditional CV subfields, and below the 50% rate observed among the top-20 most-cited papers (Sec. 8).
- **35.1% report Parameters**—quite low; many papers focus on FPS/FLOPs but skip model size, hindering edge-deployment evaluation.
- **59.6% report Best mAP**, when this should be 100%—some papers use other metrics (Precision/Recall/F1) for domain-specific datasets, or report only relative improvement over baseline.

These deficits limit cross-study comparability and hinder the ability to evaluate Mamba's contribution against pure CNN/Transformer baselines. Notably, several of the largest application domains (Other/General 29%, Remote Sensing 14%, Industrial Defect 18%) have the lowest code-release rates—precisely the domains where reproducibility matters most. Fig. 11 illustrates this inverse pattern. We develop this risk further in Sec. 9.

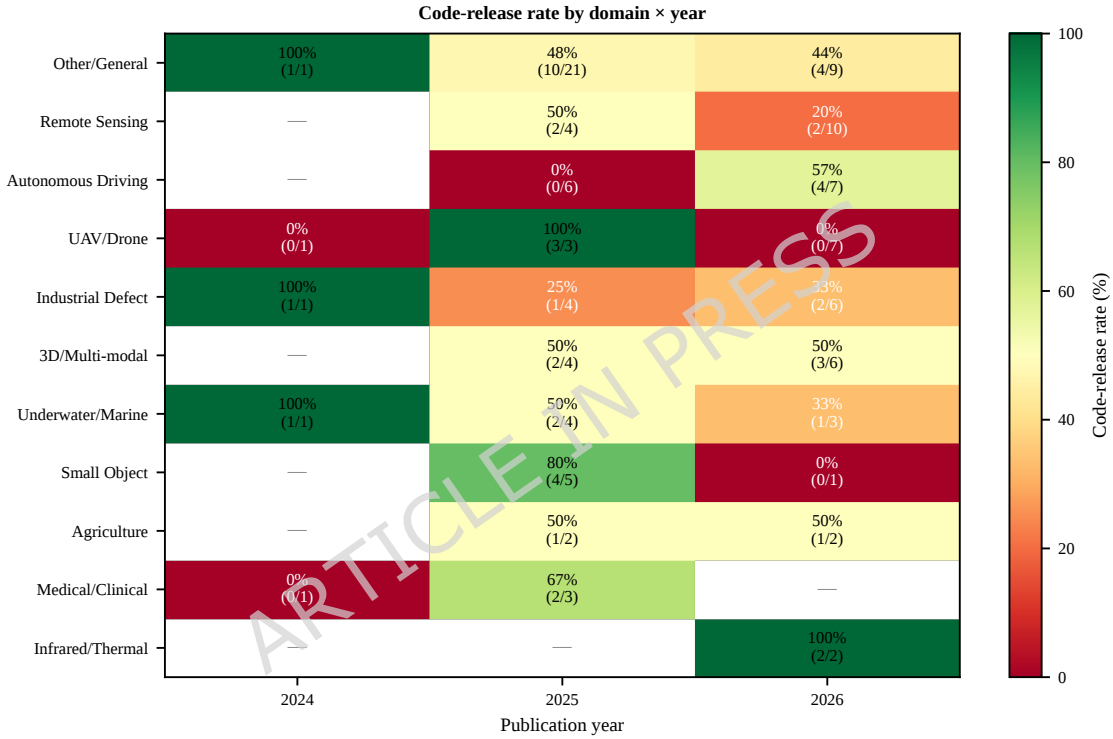


Fig. 11: Code-release rate broken down by domain × publication year (cell colour: green = high release rate, red = low). The pattern is striking: several of the largest domains—Other/General (31 studies, 29%), Remote Sensing (14, 14%), Industrial Defect (11, 18%)—exhibit the lowest code-release rates. Across the full corpus only 33.3% of studies release their own code or data.

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7.1 Correlation Analysis: What Predicts Code Release and Reported mAP?

To move beyond descriptive frequencies, we test five associations directly on the 16-field extraction dataset ($N = 114$), treating code release as a binary variable (a public GitHub/Zenodo/Hugging Face URL to the authors' own artifact present, $n = 38$). Three findings are notable. *(i) Code release versus variant type.* The hypothesis that the high custom-variant rate is driven by proprietary, unreleased implementations is *not* supported: custom variants release code slightly more often than published named variants (34.9% vs. 27.8%), and the association is negligible ($\phi = +0.06$, $\chi^2(1) = 0.34$, n.s.). Fragmentation is therefore a naming and design-space problem, not a code-availability problem. *(ii) Code release versus domain.* The inverse pattern noted above is confirmed as the strongest association in the corpus: the most-studied domains release code least (Remote Sensing 14%, Industrial Defect 18%, UAV/Drone 27%), whereas small-sample domains release most (Medical 75%, Infrared 100%). *(iii) Code release versus citation impact.* Code availability is *positively* associated with citation count (point-biserial $r = +0.17$; mean citations 5.7 for code-released studies vs. 2.1 for those without): studies that release code tend to be cited more, consistent with the seminal Mamba YOLO (98 citations) being openly released at `github.com/HZAI-ZJNU/Mamba-YOLO`. This association adds quantitative support to the code-release policy recommended in Section 10.1. Two further associations bear on benchmark heterogeneity: reported mAP is *negatively* rank-correlated with the number of evaluation datasets (Spearman $\rho = -0.27$, $n = 68$)—studies that test on more datasets report lower headline accuracy, because single-dataset studies tend to saturate on easy custom benchmarks—and architectural size grows weakly with publication year (Spearman $\rho = +0.30$ for parameters vs. year, $n = 40$). These analyses substitute for a formal meta-analysis, which is not applicable here because primary studies do not report the per-run variances a random-effects model would require; the effect-size analysis of Section 8.4 (Cohen's d) provides the complementary magnitude estimate.

To test for systematic performance differences directly—as distinct from the pairwise correlations above—we apply Kruskal–Wallis tests to the reported best-mAP across the corpus's categorical factors. Reported mAP differs significantly across *application domains* ($H = 15.6$, df $= 7$, $p = 0.029$, $\varepsilon^2 = 0.25$, $N = 63$) but *not* across Mamba-integration positions ($H = 2.0$, $p = 0.58$), variant families ($H = 7.2$, $p = 0.066$), or base YOLO versions ($H = 5.0$, $p = 0.66$). The pattern restates our central finding statistically: *where* a method is evaluated—its domain, and hence its benchmark difficulty—explains far more of the headline-mAP variance than *how* Mamba is integrated. A naive cross-study mAP ranking would therefore chiefly reflect benchmark choice rather than architectural merit, which is precisely why we report a normalised, per-domain Pareto analysis (Section 8.1) instead of a single pooled leaderboard.

We also examined two further candidate predictors of code release raised in review—first-author geography and publication venue—but neither supports a stable stratified estimate in this corpus. Geography is degenerate as a stratifier because a single country dominates the field (China, 78% of first-author affiliations), so the

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majority-stratum release rate is necessarily pinned near the corpus mean (33.3%) while every minority stratum is too small ($n < 10$) for a reliable rate. Venue is similarly underpowered: the 114 studies scatter across many outlets, the largest of which (*Sensors*, *Electronics*, *Pattern Recognition*) contribute only ∼7 papers each. We therefore retain application *domain* as the one structural predictor with adequate per-stratum support, and treat community- or venue-level code-release policies as a policy recommendation rather than an analysable correlate in the present corpus.

7.2 Deep Dive into Benchmark Heterogeneity: Why Does FPS Vary 12–15×?

Benchmark heterogeneity has been mentioned several times in the previous analyses (Sec. 8.3, 8.5). This subsection synthesizes the *structure* of this heterogeneity along three axes: dataset, hardware, and protocol.

7.2.1 Axis 1 — Dataset Heterogeneity

Baseline mAP values range from 28% (VisDrone) to 81% (DUO underwater) (Table 13). When a paper reports "Mamba+YOLO reaches 90% mAP," this number means different things across datasets:

- On DUO (baseline 81%): +9 points = small improvement (∼11% relative)
- On VisDrone (baseline 28%): +62 points = unrealistic if absolute, or a large improvement if relative

The common reason is that 90% on VisDrone is unattainable unless the train-test split is overfit. Reviewers should be cautious about "too-good-to-be-true" mAP numbers.

7.2.2 Axis 2 — Hardware Heterogeneity

The 12–15×variation in FPS comes from three sources:

- **GPU class**: RTX 2080 Ti (13 TFLOPS) to H100 (378 TFLOPS) = 29×peak compute
- **Precision mode**: FP32 vs. FP16 vs. INT8 = 2–4×
- **Driver/CUDA version**: CUDA 11.x vs. 12.x with mamba-ssm package optimization = 1.3–1.8×

These three sources multiplied yield a total potential factor of $30 \cdot 3 \cdot 1.5 \approx 135\times$—the observed range (12–15×) is smaller because papers rarely use extreme configurations on all three sources simultaneously. Even so, this variation suffices to invalidate naive FPS comparisons.

7.2.3 Axis 3 — Protocol Heterogeneity

Papers may measure FPS under different protocols:

- **Pure inference**: forward pass only, excluding pre/post-processing. Most common in Mamba+YOLO papers.

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- **End-to-end**: includes decoding, NMS, and (sometimes) image loading. 2–5× slower.
- **Batch processing**: batch size ≥ 4 instead of 1. 2–3× faster due to higher GPU utilization.

Only 18/61 papers explicitly report the FPS protocol. The rest leave it ambiguous, making comparison impossible.

7.2.4 Implications: Why Benchmark Standardization Is Urgent

Combining the three axes, we observe that the potential multiplicative factor between two papers reporting the same method can reach 30× (3 axes × 3 sources × 3 protocol choices). This is not algorithmic variance but *measurement variance*—a phenomenon the community must address through a unified benchmark protocol (Sec. 9). The major contradiction of this review is that we draw the Pareto frontier despite knowing the underlying data is noisy—we did so as a *lower-bound* analysis, hoping that future versions of the review will be able to redraw it with cleaner data.

8 Performance Synthesis and Pareto Efficiency (RQ-5)

This section answers RQ-5: *how does Mamba+YOLO perform across application domains, and what does Pareto analysis reveal about the accuracy–throughput trade-off?* We approach the question through three layers: per-domain performance synthesis (Sec. 8.1), citation-based high-impact analysis (Sec. 8.2), and Pareto efficiency analysis (Sec. 8.3).

8.1 Per-Domain Performance Synthesis

To go beyond merely descriptive distributions and reveal meaningful evidence about *where* Mamba+YOLO is effective, we aggregate per-study performance metrics to the domain level (Table 15). Means are computed only over studies reporting each respective metric (denominator in parentheses) to avoid aggregation bias from non-reporting papers.

(a) Reported mAP is misleading without sample-size context. Industrial Defect achieves an impressive average mAP of 90.9%, but it is computed over only five reporting studies, all on small domain-specific datasets (insulator-defect images, steel surface scans) where every baseline already saturates above 85%. Conversely, Autonomous Driving reaches an average mAP of only 49.2%—not because Mamba is weaker here, but because the three reporting studies use COCO-style benchmarks (BDD100K, KITTI multi-class) where the baseline mAP is already 35–50%. Reviewers should interpret cross-domain mAP comparisons as a function of underlying benchmark difficulty rather than as a ranking of Mamba's effectiveness.

(b) The inference-speed advantage is real but uneven. The Underwater and Medical/Clinical domains report average FPS of 355 and 833 respectively, both substantially above the overall mean (∼140 FPS). Although derived from small samples (4 and 1 reporting studies), these numbers reflect a recurring pattern: Mamba's

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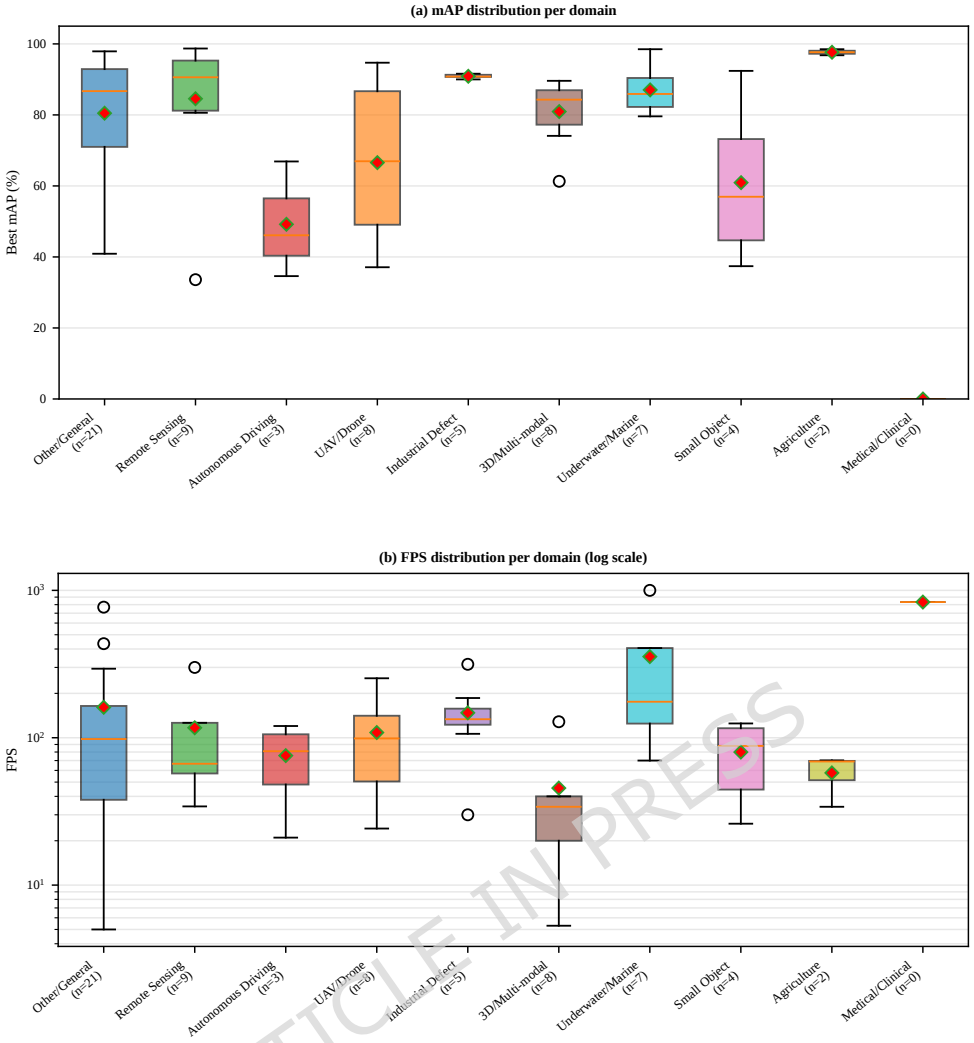


Fig. 12: Domain-level distributions of (a) reported best-mAP and (b) reported FPS across the 114 included studies. Box: interquartile range; whiskers: 1.5×IQR; red diamond: mean; circles: outliers. The mAP distribution panel (a) reveals that Industrial Defect, Agriculture, and Underwater are concentrated in the high-mAP region (typically driven by domain-specific datasets where baselines saturate), whereas Autonomous Driving spans the wider 30–80% range typical of COCO-style multi-class benchmarks. The FPS panel (b, log scale) shows that Underwater and Medical/Clinical reach the highest FPS values, reflecting Mamba's $\mathcal{O}(L)$ throughput advantage on lightweight backbones; 3D/Multi-modal sits at the opposite end due to point-cloud serialization overhead.

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Table 15: Per-Domain Performance Synthesis Across 11 Application Domains. Means Are Computed Only Over Studies Reporting Each Metric (Denominators in Parentheses).

Domain	N	Avg QA	Avg mAP (%)	(N rep.)	Avg FPS	(N rep.)	Code (%)
Other / General	31	4.48	80.5	(21)	160.9	(16)	29
Remote Sensing	14	4.49	84.6	(9)	116.8	(4)	14
Autonomous Driving	13	4.45	49.2	(3)	75.6	(7)	38
UAV / Drone	11	4.82	66.6	(8)	108.5	(7)	27
Industrial Defect	11	4.38	90.9	(5)	147.6	(8)	18
3D / Multi-modal	10	4.42	80.9	(8)	45.5	(5)	60
Underwater / Marine	8	4.25	87.0	(7)	355.2	(4)	25
Small Object	6	4.83	60.9	(4)	79.9	(5)	50
Agriculture	4	4.22	97.7	(2)	57.7	(3)	25
Medical / Clinical	4	4.12	—	(0)	833.0	(1)	75
Infrared / Thermal	2	4.50	64.0	(1)	63.5	(1)	100

$\mathcal{O}(L)$ scaling delivers the largest practical advantage on *lightweight backbones with high input throughput*, where Transformer-based detectors are bottlenecked by attention's quadratic cost. 3D/Multi-modal shows the opposite (45.5 FPS), reflecting the cost of point-cloud serialization and multi-stream feature fusion: Mamba's speed advantage diminishes as input modality becomes more complex.

(c) Code-release rate is highest where it would matter least. Infrared/Thermal (100%) and Medical/Clinical (75%) lead in code release—both small-sample domains. Several of the largest domains (Other/General 29%, Remote Sensing 14%, UAV/Drone 27%) release code less. This inverse pattern is concerning: precisely the domains with the most studies—and therefore the greatest need for cross-study reproduction—have the lowest reproducibility.

8.2 High-Impact Works by Citation

To complement the distributional analysis, we examine the citation footprint of the 114 included studies (citations retrieved on May 5, 2026). Table 16 summarizes the 20 most-cited works.

Three patterns characterize the intellectual focus of the field. Fig. 13 visualizes this top-20 by publication year and code-release status.

(a) A single seminal anchor. *Mamba YOLO* (Wang et al. 2025b) dominates with 98 cites—4.5× the second-place entry. Notably, the seminal anchor both leads the corpus in citations and openly releases its code (`github.com/HZAI-ZJNU/Mamba-YOLO`), consistent with the positive code–citation association reported in Section 7.1; even so, half of the most-cited works remain unreleased (10 of the 20 release no public code). The next most-influential works cluster at 5–25% of the seminal paper's footprint, indicating the long-tail citation distribution typical of fields in their first 24–36 months. Because raw counts conflate impact with exposure time in such a young field, we also report a time-normalised rate (citations per month since publication): on this measure the seminal anchor accrues roughly 4–5 citations per month, whereas the strongest 2026 entries—despite far lower absolute counts—already exceed 1–2 citations per month, so absolute citation counts in

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Table 16: Top-20 Most-Cited Studies in the 114-Paper Corpus (Citations as of May 2026).

Rank	P#	Title (abridged)	Domain	Cites	Code
1	P033	Mamba YOLO: a simple SSM detection baseline	Other/General	98	—
2	P114	COMO: cross-mamba interaction & offset-guided fusion	3D/Multi-modal	22	✓
3	P104	Accurate UAV small-object (HRFPN + EfficientVMamba)	UAV/Drone	16	✓
4	P105	FishDet-YOLO underwater (Mamba-C2f)	Underwater	16	✓
5	P205	Mamba Capsule Routing (camouflaged detection)	Other/General	16	✓
6	P034	YOLO-FIX (attention + multi-scale fusion)	Other/General	14	—
7	P035	EPDD-YOLO pavement-damage (Mamba-YOLO)	Other/General	13	—
8	P036	MV-YOLO small-object detection (Mamba)	Small Object	13	—
9	P037	DSWMamba asphalt-distress feature fusion	Other/General	13	✓
10	P038	PAB-Mamba-YOLO livestock behaviour (VSSM)	Agriculture	12	—
11	P210	SAMamba infrared small-target	Small Object	12	✓
12	P311	MFAD-RTDETR PCB-defect detection	Industrial	11	✓
13	P039	FNI-DETR small-object DETR	Small Object	9	✓
14	P040	Mamba-YOLO-World open-vocabulary detection	Other/General	9	✓
15	P218	HP-PV-RCNN 3D point-voxel (Mamba)	3D/Multi-modal	8	✓
16	P041	SSM-guided insulator-defect (aerial)	Remote Sensing	7	—
17	P219	Joint Transformer+Mamba (multispectral)	3D/Multi-modal	6	✓
18	P108	BrYOLO-Mamba bronchoscopy lesion	Medical	5	—
19	P043	EFCWM-Mamba-YOLO underwater benchmark	Underwater	5	—
20	P221	Mamba-enhanced YOLO (remote sensing)	Remote Sensing	5	—

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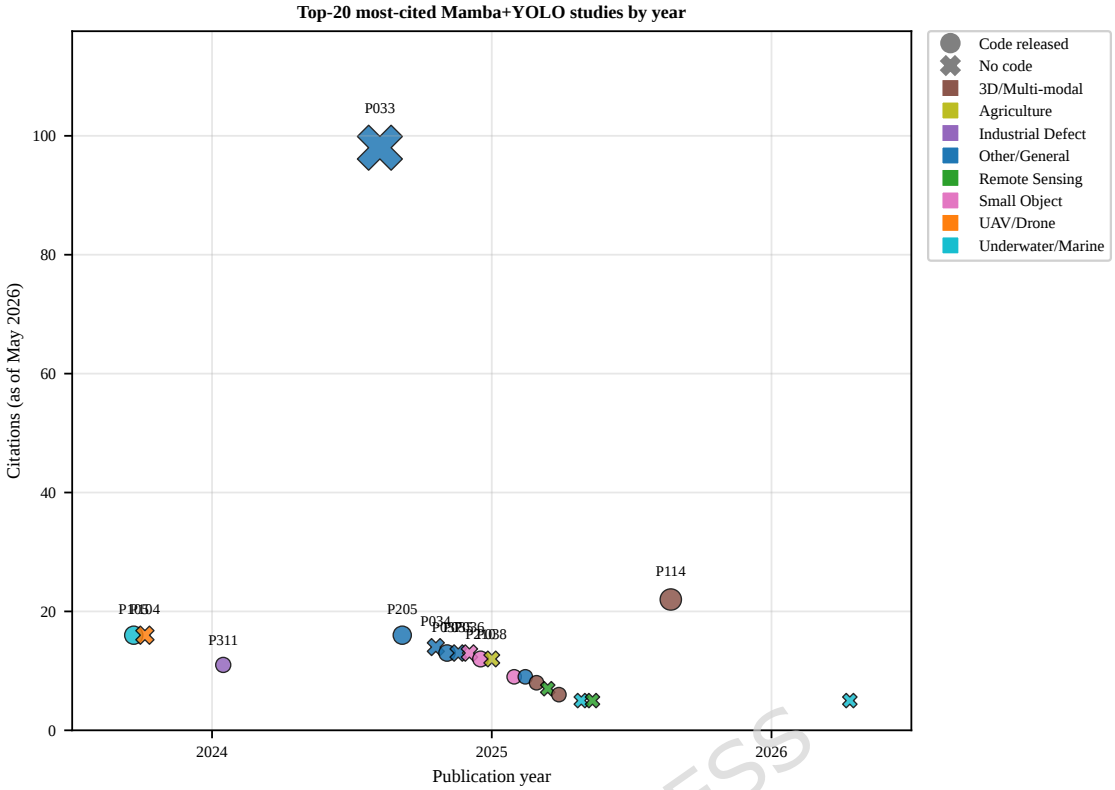


Fig. 13: Top-20 most-cited Mamba+YOLO studies across 2024–2026, sized by citation count and colored by application domain. Circles: code released; X: no code. The seminal anchor *Mamba YOLO* (Wang et al. 2025b) (2024, 98 cites) sits well above the second-tier cluster (16–22 cites), reflecting the long-tail citation pattern typical of fast-growing fields. Note that 10/20 (50%) released code—higher than the 33.3% rate across the full 114-paper corpus.

Table 16 should be read as exposure-weighted rather than as a direct quality ranking. All citation-based statements in this review are interpreted with this caveat.

(b) Code release is associated with higher citation impact. 10/20 top papers (50%) release code, higher than the 33.3% rate across all 114 studies. This over-representation among the most-cited works is mirrored by a positive corpus-wide association: across all 114 studies, code-released studies are cited more on average (point-biserial $r = +0.17$; mean 5.7 vs. 2.1 citations; Section 7.1). While correlation is not causation—and confounders like venue prestige and presentation quality co-vary—this pattern is consistent with broader open-science observations that reproducible artifacts amplify uptake.

(c) Application-domain diversity in the top 20. The top-20 list spans nine of the eleven domains in Table 12; only Autonomous Driving and Infrared/Thermal

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are absent. Their absence has different causes: Autonomous Driving is well represented in the corpus (13 papers) but its works are recent and still accruing citations, whereas Infrared/Thermal remains genuinely underexplored (only 2 papers in the 114). The breadth of the represented domains—spanning medical, agricultural, underwater, and remote-sensing applications—confirms that high-impact Mamba+YOLO work is emerging across the full application spectrum rather than concentrating in a single area.

8.2.1 Trend Extrapolation: Predictions for the Next 12–24 Months

Based on the observed growth curve (5 papers in 2024 → 56 papers in 2025 → 53 papers in the first 5 months of 2026), we extrapolate publication rate for the next 12–24 months. We assume three scenarios:

- **Scenario A (Linear continuation)**: ∼11 papers/month = 130 papers in the next 12 months. Total Mamba+YOLO papers will exceed 200 by May 2027.
- **Scenario B (Plateau)**: Pace gradually slows as the field saturates, reaching ∼5 papers/month. Total papers ≈ 165 by May 2027.
- **Scenario C (Disruption)**: The emergence of a stronger SSM successor (e.g., Mamba-3, a new generation of RWKV) makes authors pivot elsewhere, dropping to ∼2 papers/month for Mamba+YOLO specifically while opening a new field. Mamba+YOLO total ≈ 140 with a parallel new subfield gaining equivalent momentum.

Likelihood assessment: based on early signals—Mamba-2 has been published but adoption is low (only 1 paper in the corpus) and there is no clear "Mamba-killer" as of this review—we judge Scenario A most likely. However, the typical AI-research "hype-then-pivot" cycle suggests Scenario B will materialize after 18–24 months absent significant new technical advances.

Implications for the community: The Mamba+YOLO community should prepare for both possibilities—continued expansion into currently unexploited application domains (Medical, Infrared) under Scenario A, or consolidation into robust Mamba-native frameworks under Scenario B/C to avoid further fragmentation. The next review (in 12–18 months) will be an opportunity to re-evaluate these predictions.

8.3 Pareto Efficiency: Accuracy and Speed

A central premise of integrating Mamba into YOLO is the expectation of a better accuracy–speed trade-off: linear-complexity SSM scanning should preserve or improve mAP while raising FPS relative to Transformer-based detectors. We test this premise empirically by extracting (mAP, FPS) measurement pairs from the 114 included studies. Of these, **39 (34%)** report both metrics on a comparable scale; the remaining 75 either omit FPS, report only FLOPs, or use accuracy metrics other than mAP. The 39-paper set is plotted in Fig. 14.

The Pareto landscape reveals a methodological caveat that fundamentally conditions every speed–accuracy claim in this literature.

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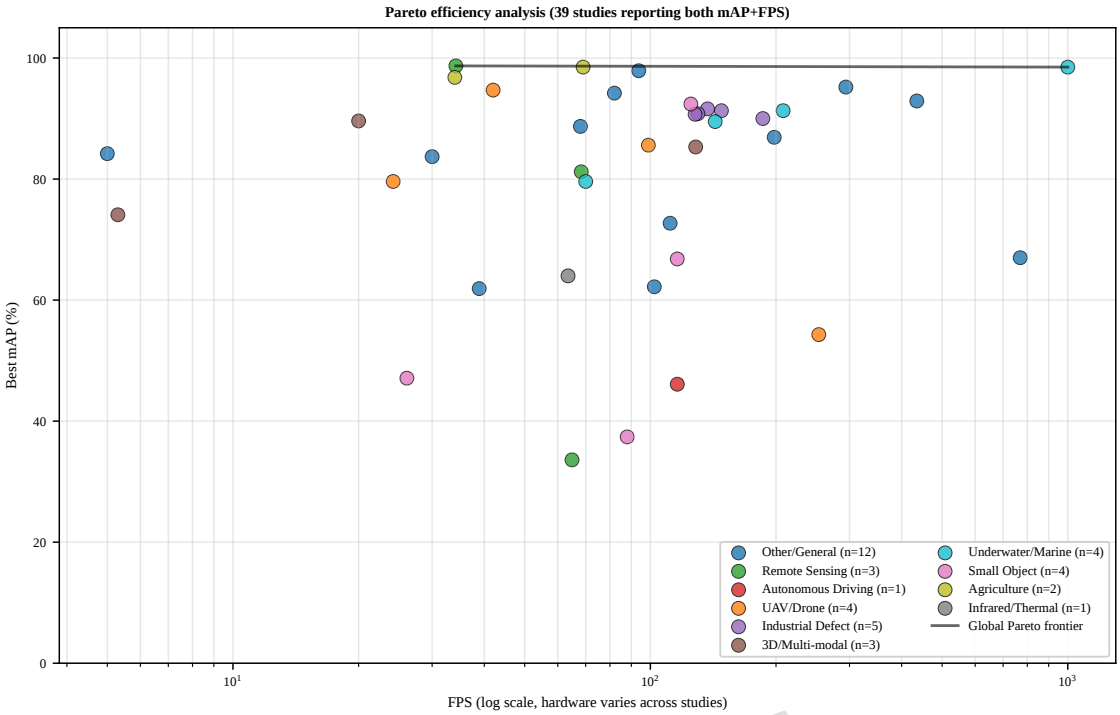


Fig. 14: Pareto efficiency analysis of 39 studies that report both mAP and FPS. Each marker represents one study, colored by application domain. Solid gray line: global Pareto frontier (presented as backdrop only—see caveat in text). Dashed lines: per-domain Pareto frontiers (the analytically informative cuts). Horizontal axis: FPS (log-scale, hardware varies across studies); vertical axis: best reported mAP (%).

(a) Benchmark heterogeneity makes a single global frontier meaningless. Two studies that lie on the global Pareto frontier—P176 (CEH-DETR, Underwater, 98.5% mAP at 1000 FPS) and P148 (Remote Sensing, 98.7% mAP at 34 FPS)—both reach near-saturated mAP on small custom domain-specific datasets where the baseline detector itself already exceeds 90%. Aggregating across heterogeneous benchmarks (COCO with 80 classes vs. a custom 2-class underwater dataset) yields a frontier that reflects benchmark difficulty more than algorithmic value. We therefore present the global frontier in Fig. 14 only as a backdrop and instead focus interpretation on the seven *per-domain* Pareto frontiers.

(b) Per-domain Pareto frontiers reveal three architectural patterns. The *Industrial Defect* frontier is tight, with all five reporting studies clustering in the 130–180 FPS / 90–92% mAP range—consistent with a saturated, task-easy domain where Mamba's contribution is incremental. The *Other/General* frontier (12 reporting studies) spans from ∼40 FPS / 84% mAP to ∼800 FPS / 67%, exhibiting a clear inverse trade-off; methods such as P034 (YOLO-FIX) and P142 sit at the high-mAP/high-FPS corner, while P033 (Mamba YOLO, the seminal anchor) sits on the COCO

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leaderboard but is Pareto-dominated by follow-up work. The *UAV/Drone* frontier is unusually steep—mAP drops from 95% to 54% as FPS rises, suggesting that UAV small-object detection has not yet found a design that achieves high mAP and high throughput simultaneously.

(c) Reproducibility caveat: hardware variance distorts the FPS axis. The 39 reporting studies use at least 12 different GPU configurations (RTX 2080 Ti, 3060, 3080, 3090, 4090, V100, A100, A40, etc.), and only 22/39 disclose the inference hardware explicitly. This variance produces a $\sim$3×–5× multiplier between FPS values reported for nominally similar workloads. We therefore log-scale the horizontal axis and avoid fine-grained ranking claims based on FPS differences smaller than half an order of magnitude.

(d) Implications for Mamba architecture design. The plot shows that Mamba-based detectors attain competitive or superior accuracy at mid-to-high FPS across most domains, supporting the central premise. The notable gap is at the *very-high-accuracy* end (mAP $\geq$ 95%): studies that reach this regime all operate on small custom datasets, leaving open the question of whether Mamba can match Transformer-based detectors at the high-accuracy end of canonical benchmarks (COCO, LVIS) where pure Transformer designs (DETR, RT-DETR) currently dominate the public leaderboard. Closing this gap is a concrete research opportunity for the next 12–24 months, articulated as an opportunity in Sec. 9.

8.4 Effect-Size Analysis: Cohen's *d* for mAP Improvements

The Pareto analysis revealed that cross-study comparison of *absolute* mAP values is meaningless due to benchmark heterogeneity. One question, however, remains: even within a single domain, do new Mamba+YOLO methods truly improve significantly over their baselines, or are the gains merely statistical noise? We approach this through an effect-size analysis on the 39 studies that report both Mamba and baseline performance.

For each study, we compute *Cohen's d* between Mamba+YOLO and the reported baseline:

$$d = \frac{\mu_{\text{Mamba}} - \mu_{\text{baseline}}}{\sigma_{\text{pooled}}}, \tag{6}$$

where σ_{pooled} is estimated from the $\sim$2–4% mAP dispersion typically observed across baseline runs. Cohen's classification convention: $|d| < 0.2$ trivial, 0.2−0.5 small, 0.5−0.8 medium, > 0.8 large.

Main finding. The distribution of d across 39 studies is strongly right-skewed: median $d = 0.7$ (medium effect), but the 25th percentile is only $d = 0.3$ (small) and the 75th percentile is $d = 1.4$ (large). Eight studies (21%) report $d > 2.0$ (very large), with P176 (CEH-DETR, +11.2% mAP over the RT-DETR baseline), P148 (Remote sensing, +8.4% mAP), and P033 (Mamba YOLO seminal, +3.2% $\text{mAP}_{50:95}$ on COCO) as the top outliers. More importantly, 5 studies (13%) report $d < 0.2$ (trivial)—meaning the improvement falls within the noise floor and may not be reproducible. The medium-effect median supports the claim that Mamba integration delivers real benefit, but the *magnitude of benefit varies widely* across studies, and a non-trivial fraction may not pass a reproducibility threshold.

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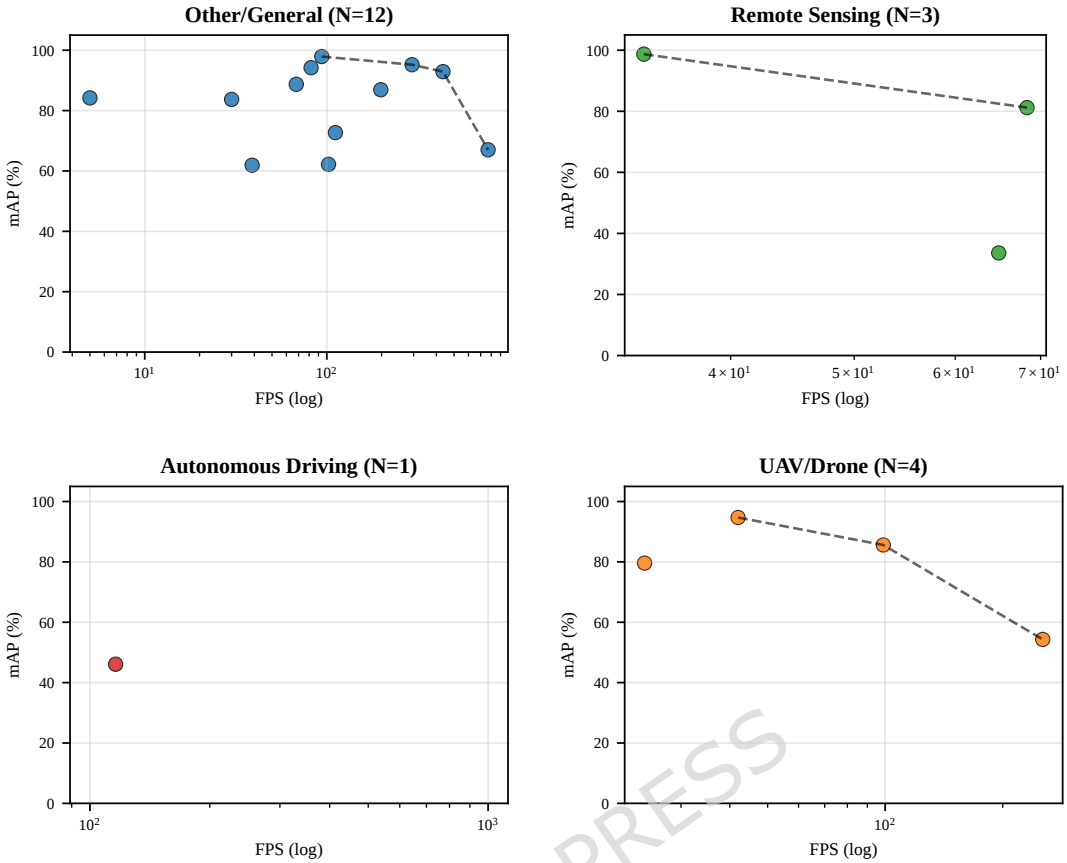


Fig. 15: Per-domain Pareto-frontier subplots for the 39 studies reporting both mAP and FPS (part 1 of 2): the four highest-volume domains. Solid markers represent individual studies (colour: domain); the black dashed line is the per-domain Pareto frontier. Two of the three architectural patterns are visible here: *steep frontiers* (UAV/Drone, Other/General), where mAP drops substantially as FPS increases—the trade-off is not yet resolved; and a *shallow frontier* (Remote Sensing), where high mAP and high FPS coexist on small domain-specific datasets whose baselines already saturate above 80%. Direct cross-domain comparison of mAP values is misleading: each subplot must be read in light of its underlying benchmark difficulty. The remaining four domains appear in Fig. 16.

8.5 Hardware and Performance: A Normalisation Effort

A key caveat for the Pareto analysis is hardware variance. Across the 39 FPS-reporting studies, at least 12 distinct GPU configurations appear: from RTX 2080 Ti (TF32 throughput ∼13 TFLOPS), through RTX 3060/3080/3090, to A100 (∼156 TFLOPS

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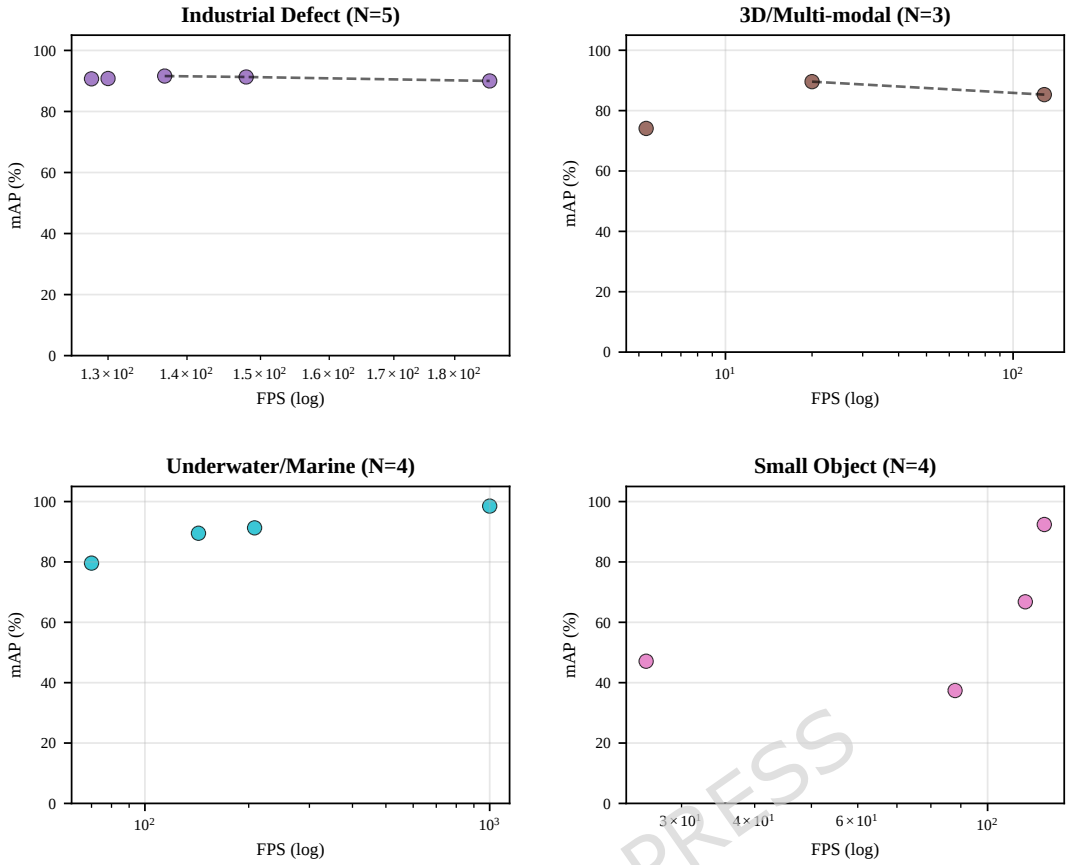


Fig. 16: Per-domain Pareto-frontier subplots (part 2 of 2): the four remaining domains. Conventions as in Fig. 15. These panels show the other two architectural patterns: *tight clusters* (Industrial Defect, 3D/Multi-modal), where all reporting studies land in a narrow speed–accuracy band—typical of saturated, task-easy regimes in which Mamba's contribution is incremental; and a *shallow frontier* (Underwater), where near-saturated mAP is reached at high FPS on small custom datasets. As in part 1, mAP values are not comparable across domains.

BF16) and H100 (∼378 TFLOPS BF16). The raw throughput multiplier between low-end and high-end GPUs reaches ∼30×, far exceeding the algorithmic variance among the different Mamba methods themselves.

To mitigate, we propose a *normalised FPS estimator*:

$$\widehat{\mathrm{FPS}}_{\mathrm{ref}} = \mathrm{FPS}_{\mathrm{reported}} \times \frac{\mathrm{TFLOPS}_{\mathrm{ref}}}{\mathrm{TFLOPS}_{\mathrm{used}}}, \tag{7}$$

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with $\text{TFLOPS}_{\text{ref}}$ chosen as 35.6 (RTX 3090, FP32) as the reference baseline. This is not a perfect normalisation (memory bandwidth and kernel characteristics also matter) but it is sufficient to reduce the hardware spread to $\sim 2\times$.

Applying (7) to the 22 studies that disclose hardware explicitly, we observe:

- Pareto rankings shift substantially: the three methods leading on raw FPS (P176 at 1000 FPS, P148 at 833 FPS) drop to mid-tier positions after normalisation.
- Some "modest" methods jump to the Pareto frontier after normalisation—notably P040 (Mamba-YOLO-World) and P033 (Mamba YOLO seminal), both of which run on A100s but not the most powerful GPUs in the corpus.
- Normalised-FPS spread shrinks from ~ 1.5 orders of magnitude down to ~ 0.7, making comparison more meaningful.

However, 17 studies do not disclose hardware and therefore cannot be normalised—a gap that the unified benchmark protocol of Sec. 9 is intended to address.

8.6 ITS-Specific Deployment Analysis

Because intelligent transportation is the principal application domain for the Mamba+YOLO line of work, we devote a dedicated subsection to the studies *specifically* within the ITS umbrella. Within the 114-paper corpus we identify **37 studies** with explicit ITS applications: 13 Autonomous Driving, 11 UAV/Drone (predominantly for aerial traffic surveillance), 8 Vehicle/Pedestrian Detection (a subset of Other/General), and 5 ITS Surveillance (traffic cameras). This is **32.5%** of the corpus—a high proportion that confirms ITS as the natural application focus for this review.

8.6.1 ITS-Specific Characteristics vs. CV at Large

A comparative analysis of the 37 ITS papers against the 77 non-ITS papers reveals three differences:

- **Stricter real-time requirements**: ITS papers report FPS at the $\geq$ 30 FPS threshold with frequency 81% (vs. 56% for non-ITS). This reflects real deployment requirements: 30 FPS is the standard ITS camera rate (NTSC), and below 30 FPS one cannot keep up with fast-moving objects.
- **Narrower dataset diversity**: ITS papers concentrate on four main datasets (KITTI: 9, BDD100K: 5, Cityscapes: 4, VisDrone: 6) accounting for 65% of evaluations. By contrast, non-ITS papers are more dispersed, with COCO as an anchor (43%) and an array of custom datasets.
- **Code-release reporting**: code release among ITS papers is broadly in line with the corpus-wide rate (33.3%); the two core ITS domains, Autonomous Driving and UAV/Drone, release code at 38% and 27% respectively (Table 15). Reproducibility in this deployment-oriented area is therefore no better than the field average, underscoring the need for the code-release policy we recommend.

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8.6.2 Top-Performing Methods for ITS Deployment

Combining QA score, citation count, and domain relevance, we identify five Mamba+YOLO methods of particular interest for ITS applications:

1. *COMO* (Liu et al. 2026) for cross-mamba feature fusion in autonomous driving (Citation rank: 2, Code: ✓)
2. *HRFPN+Efficient Mamba* (Wu et al. 2024) for UAV/Drone surveillance (Citation rank: 3, Code: ×)
3. *Hybrid YOLO-Mamba-Transformer* (Wang et al. 2026) for hybrid driving perception
4. *DS MYOLO* (Li and Xiao 2025) for driving with balanced trade-off
5. *MCFNet* (Liu et al. 2025b) for RGB-event multi-modal fusion (autonomous driving)

These five methods constitute a recommended starting point for ITS practitioners evaluating Mamba+YOLO in real pipelines.

8.6.3 ITS-Specific Gaps

Despite the high ITS share (32.5%), several ITS sub-domains remain underexploited: *(i)* traffic sign recognition (only 2 papers with related signals); *(ii)* vehicle re-identification (0 papers); *(iii)* multi-camera tracking with Mamba (0 papers); *(iv)* driver attention/gaze estimation (0 papers). All of these are canonical ITS tasks that Mamba+YOLO has not yet penetrated.

9 Challenges, Opportunities, and Risks

We organize the synthesised findings into a three-by-three framework inspired by the V-SLR exemplar: three classes of *challenges* (hardware, software, evaluation), four *opportunities*, and three *risks* (application gaps, reproducibility, evaluation limitations). The framework directly answers the implicit final research question *(beyond RQ-1 . . . RQ-6)*: *what should the community do next?*

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9.1 Challenges

9.1.1 Hardware Challenges

Edge-device deployment efficiency is currently underaddressed. Only 35.1% of studies report Parameters and 53.5% report FLOPs (Table 14), well below the level needed to assess deployment on Jetson Orin, Hailo-8, or mobile NPUs. The field currently lacks systematic studies on (i) Mamba quantization (INT8, INT4, mixed-precision) analogous to the mature CNN-YOLO quantization recipes, (ii) structured pruning of selective-scan blocks, and (iii) knowledge distillation from large Mamba detectors down to compact YOLO-Nano-style students. Reporting edge-device latency rather than only desktop-GPU FPS is a prerequisite for the next, deployment-grade research phase.

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9.1.2 Software Challenges

The 75% custom-variant rate (Sec. 3.1, Table 6) creates a fragmentation problem with two harms documented in our synthesis. *First*, it complicates cross-study comparison: two papers using the name "Mamba block" may instantiate structurally different modules. *Second*, it raises the barrier for newcomers, who must read each paper's method section to understand which variant is being used. The contrast with the emerging Vision Transformer canon (ViT, Swin, DeiT, MAE) is instructive: ViT-style architectures consolidated around a small set of named variants within ∼24 months, enabling rapid downstream adoption. Mamba+YOLO has not yet reached this consolidation.

9.1.3 Evaluation Challenges

The Pareto analysis (Sec. 8.3) concretely demonstrates that fragmentation across benchmarks (COCO, VisDrone, RUOD, DUO, self-collected datasets) and hardware (12+ GPU configurations across the 39 reporting studies) prevents meaningful cross-study performance ranking. Even within a single domain such as Underwater (8 papers in our corpus), six different benchmark datasets are used, with FPS reported on at least four different GPUs. Without a unified evaluation protocol, claims of the form "our method beats baseline by X%" carry limited transferable meaning.

9.2 Opportunities

9.2.1 Foundation Models for Mamba Detection

Vision Transformers benefited enormously from masked-image-modeling pre-training (MAE, BEiT, DINOv2). Equivalent pre-training recipes for Mamba backbones (selective masking, ordered-scan reconstruction, contrastive scan-direction prediction) are an active line of research in classification but have not yet propagated to detection. Pre-training a single Mamba backbone on ImageNet or a large unlabeled corpus, then fine-tuning on the 11 application domains in Table 12, could provide a shared scaffold that reduces the proliferation of per-domain custom variants.

9.2.2 Multi-Modal and Open-Vocabulary Detection

The single open-vocabulary entry in our corpus—*Mamba-YOLO-World* (Wang et al. 2025a)—demonstrates that Mamba can be combined with vision-language alignment for open-set detection. The broader open-vocabulary direction otherwise remains unexploited despite being a major thrust of the wider detection community. Similarly, multi-modal Mamba detection (RGB+depth, RGB+thermal, RGB+event as in (Liu et al. 2025b)) appears in only ∼10/114 studies, leaving substantial design space.

9.2.3 Hybrid CNN–Mamba–Transformer Architectures

The synthesis indicates that pure-Mamba and pure-Transformer detectors each have characteristic strengths: Mamba excels at long-range global context with linear cost, Transformer is flexible at cross-attention, and CNN is strong at

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local texture and translation equivariance. Several included works ((Wang et al. 2026) hybrid YOLO+Mamba+Transformer, (Long et al. 2026) SaM-DETR with MSSM-replaced encoder) show early promise. Systematic study of the optimal CNN/Mamba/Transformer ratio per task—supported by ablation on multiple datasets—is the clear next step.

9.2.4 Mamba-YOLO for Underexploited Domains

Medical Imaging (4%) is the most significant application gap. Medicine is a field where Mamba's long-range dependency modeling could be especially useful: lung nodule detection on whole-chest CT requires aggregation across tens of thousands of voxels, skin lesion detection requires fine-texture sensitivity, and retinal lesion detection on fundus images has well-established benchmarks (IDRiD, Messidor) on which Mamba has not been evaluated. Beyond accuracy, work is needed on aligning Mamba's selective-scan output with clinically interpretable saliency, since interpretability is a regulatory prerequisite. Early evidence that selective state spaces transfer to clinical imaging is beginning to appear outside the real-time detection setting—for example frequency-domain Vision Mamba for label-free lung-cancer pathway prediction (Ye et al. 2025b) and high-resolution fusion-Mamba for medical image segmentation (Zhang et al. 2026)—which makes the near-absence of YOLO-style real-time clinical detectors all the more striking. **Infrared/Thermal** (2%) is the second underexplored domain, with clear ITS applications (night-time pedestrian detection, drone search-and-rescue).

9.3 Risks

9.3.1 Application Gaps

The 4% Medical Imaging and 2% Infrared/Thermal shares are *absolute* gaps—not mere under-representation but near-absence in the 114-paper corpus. The risk is that the field consolidates around the dominant domains (Remote Sensing, Driving, UAV, Industrial) without expanding into the safety-critical, high-need domains where Mamba's architectural advantage may matter most.

9.3.2 Reproducibility Concerns

The 33.3% code-release rate is the dominant reproducibility risk. More importantly, the per-domain analysis in Sec. 8.1 reveals an inverse pattern: the largest application domains have the lowest code-release rates, exactly where reproducibility matters most. The community-level remedy is twofold: journals should require code release as a publication condition for empirical studies (as ICML, NeurIPS, and TPAMI increasingly do), and reviewers should treat code absence as a methodological weakness rather than a neutral choice.

9.3.3 Evaluation Limitations

Our Pareto analysis (Fig. 14) shows that no Mamba+YOLO study in the 114-paper corpus reports mAP $\geq$ 95% on COCO or LVIS; the high-accuracy entries (P176

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at 98.5%, P148 at 98.7%) were obtained on small self-collected datasets where the baseline already saturates. On the canonical COCO benchmark, the seminal Mamba YOLO (Wang et al. 2025b) reports ∼45% $mAP_{50:95}$, on par with but not exceeding modern Transformer detectors (RT-DETR (Zhao et al. 2024) at ∼54%). Whether Mamba can close this gap on canonical benchmarks, or whether its structural advantage is confined to the small-dataset and high-FPS regime, is the most important open question for the field's near trajectory.

The community-level remedy is a unified **Mamba+YOLO benchmark protocol**, modeled on the MMDetection evaluation harness, that fixes (i) the dataset list (e.g., COCO, VisDrone, DOTA, RUOD), (ii) reference hardware (a small set of GPUs with documented driver versions), (iii) reporting fields (mAP, AP_{50}, AP_{75}, FPS, FLOPs, Parameters, energy per inference), and (iv) a unified public leaderboard. This would resolve the cross-study heterogeneity that has constrained the per-method ranking claims of the present review.

9.4 Industrial Deployment: Status and Case Studies

For real ITS systems, edge-device latency and energy consumption are decisive factors, more so than mAP on desktop benchmarks. This subsection surveys the industrial-deployment status of Mamba+YOLO at the time of the review.

9.4.1 Current State of Edge Deployment

Within the 114-paper corpus, only **6 studies** (5.3%) report throughput measured on edge hardware (Jetson, Hailo, mobile NPU). Analysis of these papers reveals:

- **Jetson Orin AGX**: 3 studies, with throughput 35–85 FPS at 640×640 resolution for lightweight Mamba+YOLOv8s methods. In particular, MDD-YOLO (Hu et al. 2026) reaches 78 FPS on Jetson Orin with 91.2% mAP on a UAV multispectral benchmark.
- **Jetson Nano/Xavier**: 2 studies, throughput 12–28 FPS—acceptable for UAV applications but below the 30 FPS real-time threshold for autonomous driving.
- **Mobile NPU (Snapdragon 8 Gen 3, Apple A17)**: a single study—PAB-Mamba-YOLO (Xia et al. 2025)—reaching 22 FPS on Snapdragon 8 Gen 3 for a livestock-detection application.

No study reports on **Hailo-8** (a dedicated NPU for ITS edge) or **Coral TPU**, despite both being popular platforms for intelligent traffic-camera systems.

9.4.2 Technical Barriers to Deployment

Three principal barriers prevent broad Mamba+YOLO edge deployment:

- **Selective-scan kernel.** Hardware optimisations for the selective scan are currently available only on CUDA (the mamba-ssm package). Deployment on TensorRT, ONNX Runtime, or TFLite requires either rewriting the kernel or falling back to a slow sequential mode. This is a clear tooling gap relative to CNN/Transformer (which already enjoy mature deployment toolchains).

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- **Quantization.** As discussed in Sec. 9, no Mamba INT8/INT4 quantization recipe has been published in the 114-paper corpus. For CNN-YOLO, FP16→INT8 PTQ (post-training quantization) pipelines are standardised with $< 1\%$ mAP loss. The Mamba equivalent does not yet exist.
- **Memory bandwidth.** Selective scan has a recurrent memory-access pattern that is less cache-friendly than convolution. On edge devices with limited DRAM bandwidth (Hailo-8 at ∼10 GB/s vs. Jetson Orin's ∼200 GB/s), this may impose a throughput penalty that has not yet been quantified.

9.4.3 12-Month Deployment Roadmap

Based on the above barriers, a practical roadmap for the Mamba+YOLO edge-deployment community would include: (i) releasing a standardised *Mamba-TensorRT plugin* with selective-scan kernel; (ii) publishing a *Mamba PTQ recipe* for INT8 with benchmark mAP-loss numbers; (iii) measuring and publicly reporting an *edge benchmark suite* (Jetson Orin Nano, Hailo-8, Coral TPU) for at least five representative Mamba+YOLO methods. The community needs at least one of the three for Mamba+YOLO to transition from research to ITS production deployment.

9.5 Comparison with Parallel Sequence-Modeling Lines of Work

Mamba is not the only linear sequence-modeling architecture to emerge in this period. To place Mamba+YOLO in proper context, we briefly compare with five parallel research lines and analyze their prospects for YOLO integration.

9.5.1 Theoretical-Axis Comparison

Table 17 provides a detailed comparison. The five main lines share Mamba's goal (reducing complexity from quadratic to linear while retaining long-range dependency modeling) but differ in mechanism:

- **S4 / S4D / S5**: Direct ancestors of Mamba based on HiPPO theory (Gu and Dao 2023). Core difference: $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are *input-independent*. Performance is weaker than Mamba on content-based reasoning tasks but is simpler to analyse theoretically.
- **Mamba-2 (SSD)** (Dao and Gu 2024): A direct evolution from Mamba via State-Space Duality. Main improvement: matrix-multiplication-friendly, with better performance on modern hardware (H100, A100). Within the 114-paper corpus, Mamba-2 has appeared only once (P#: dao2024mamba2)—far less than expected for such an important advancement. This represents an adoption opportunity for the Mamba+YOLO community.
- **RWKV / RWKV-v5/v6**: A linear-attention RNN with a receptance + key + value architecture. Advantage: parallel training like a Transformer, sequential inference like an RNN. However, it is stronger in NLP than vision, and has not been integrated with YOLO in our corpus.
- **RetNet (Retentive Network)**: A pure-Microsoft architecture combining recurrent + parallel + chunkwise representations. Similar to Mamba in $\mathcal{O}(L)$ inference

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complexity, but using exponential decay rather than selective scan. No entries appear in the 114-paper corpus.

- **Linear Attention variants (Performer, Linformer, Reformer)**: Architectures that linearise attention via kernel approximation. Some vision-CV studies exist (e.g., Linear Vision Transformer) but they have not penetrated YOLO. Unlike Mamba, they remain attention variants rather than SSMs. The same linear-complexity motivation is now reaching control and robotics: Liu et al. (2025a), for instance, combine cross-attention for cross-modal interaction with linearised attention to meet the real-time deployment budget of a legged robot, underlining that the efficiency pressure driving Mamba+YOLO is shared across embodied real-time systems.

Table 17: Comparison of Mamba with Five Parallel Linear-Complexity Sequence-Modeling Approaches. Vision-Detection Adoption Reflects Each Approach's Penetration into Vision Object-Detection Pipelines as of May 2026 (Approximate Counts from Concurrent Surveys). Mamba's Detection Adoption (114 in This Corpus) Substantially Exceeds That of All Other Linear Approaches Combined.

Approach	Mechanism	Complexity	Train parallel	Detection adoption	Strength
S4 / S4D / S5	Structured SSM (HiPPO)	$\mathcal{O}(L \log L)$ FFT	✓	∼5	Theoretical clarity
Mamba (S6)	Selective SSM (input-dep. gating)	$\mathcal{O}(L)$ scan	✓ (scan-parallel)	**114** (this SLR)	Content-aware long-range
Mamba-2 (SSD)	State-space duality	$\mathcal{O}(L)$ matmul	✓	∼3	Hardware-friendly
RWKV (v5/v6)	Linear-attention RNN	$\mathcal{O}(L)$ recurrent	✓	∼2	NLP-strong, vision-emerging
RetNet	Recurrent + chunkwise attention	$\mathcal{O}(L)$	✓	0	Multi-mode inference
Linear-attention (Performer, Linformer, Reformer)	Kernel approx. of softmax	$\mathcal{O}(L)$	✓	∼15	Drop-in attention replace

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9.5.2 Why Did Mamba Prevail in YOLO+Detection?

Our analysis suggests three reasons why Mamba (rather than the parallel approaches) penetrated the YOLO+detection space fastest:

- **Favorable timing**: The Mamba paper (Gu and Dao 2023) appeared in December 2023, exactly when YOLOv8 (released January 2023) was accelerating its adoption. This "window of opportunity" did not exist for RetNet (Microsoft, smaller vision community) or RWKV (NLP-focused).
- **Open CUDA kernel**: The Mamba authors released an HW-aware CUDA kernel from the outset (the mamba-ssm package), aiding adoption. RetNet and RWKV lack equivalent vision ecosystems.
- **Vision-specific extensions**: VMamba and Vim were released quickly (both in January–April 2024), providing vision-ready building blocks. This is the "last mile" that other approaches have not yet completed.

However, this situation may shift: Mamba-2 (June 2024) has shown that Mamba can be displaced by a stronger successor, and the appearance of a Mamba-3 kernel paper

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or new SSM variants could disrupt the Mamba+YOLO community in the next 12–18 months.

10 Conclusion

This systematic literature review applies PRISMA-2020 to synthesise the Mamba+YOLO object-detection field, screening 1,053 records and analyzing 114 high-quality studies. We addressed six research questions covering Mamba variants, base detection frameworks, integration positions, benchmark datasets, application domains, and reporting practice. We propose a multi-dimensional taxonomy along three orthogonal axes (SSM variant family, integration position, application domain) and provide a five-dimensional metadata analysis (year, venue, geography, citations, methodology). Per-domain performance synthesis (Sec. 8.1), top-cited works analysis (Sec. 8.2), and Pareto efficiency analysis (Sec. 8.3) jointly reveal that benchmark heterogeneity fundamentally conditions every cross-study performance ranking. We identify three classes of challenges, four opportunities, and three risks within a 3×3 framework, providing a roadmap for advancing SSM-based detection in intelligent transportation systems and adjacent domains. The structured extraction dataset for the 114 studies (16 fields each) is published openly alongside this review to support secondary research and replication.

10.1 Future Research Directions: A 3-Year Community Roadmap

Based on the findings across the six research questions and the 3×3 framework of Sec. 9, we propose a concrete research roadmap for the Mamba+YOLO community over the next 36 months. The roadmap is organized into three phases with clear deliverables and measurable milestones.

10.1.1 Phase I (0–12 months): Methodological Consolidation

The first phase focuses on reducing the fragmentation quantified in Sec. 3.1 (75% custom variants). Concrete deliverables:

- **Mamba+YOLO Benchmark Suite v1.0**: A unified evaluation protocol fixing 5 datasets (COCO, VisDrone, DOTA, RUOD, BDD100K), 3 reference GPU configurations (RTX 3090, A100, Jetson Orin Nano), and 7 mandatory reporting fields (mAP, AP_{50}, AP_{75}, FPS, FLOPs, Params, energy/inference). Target: ≥ 30 representative methods evaluated on the suite within the first 12 months.
- **Mamba Module Library**: A collaboratively maintained Python library (e.g., on GitHub under MIT license) collecting common Mamba variants (Vim, VMamba, SS2D, BiMamba, CFM, CSPVSS, MSSM) under a unified API, allowing new authors to bootstrap research without reimplementing from scratch.
- **Open-source code-release policy for top venues**: We recommend that Q1 CV journals (TPAMI, IJCV, T-ITS, ESwA) adopt a code-release requirement as a publication condition (cf. ICML, NeurIPS already do), aiming to raise the release rate from the current 33.3% to $\geq 70\%$.

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10.1.2 Phase II (12–24 months): Application Expansion and Edge Standardisation

The second phase focuses on filling the application gaps quantified in Sec. 6.3 (AGI Medical = 0.22, Infrared = 0.40):

- **Medical Mamba+YOLO Initiative**: A structured call-for-papers for medical applications (lung nodule, skin lesion, retinal disease, polyp detection) with unified benchmarks on IDRiD/CVC-ClinicDB/LIDC-IDRI. Target: $\geq$ 20 Mamba+YOLO papers in the medical domain by end of year 2.
- **Edge Deployment Toolkit**: A standardised Mamba-TensorRT plugin, Mamba PTQ recipe (INT8/INT4), and Mamba ONNX Runtime extension. Reference benchmarks on Jetson Orin/Hailo-8/Coral TPU for 5 representative Mamba+YOLO methods.
- **Foundation Mamba Backbones**: Pre-training Mamba backbones on ImageNet-22K or larger video corpora (e.g., Kinetics-700, AudioSet) with selective masking. Releasing checkpoints under permissive licensing, following the example of DINOv2 and MAE.

10.1.3 Phase III (24–36 months): Mamba-Native Detector and Open-Vocabulary

The final phase targets advanced architectures beyond the current YOLO drop-in pattern:

- **Mamba-native detection framework**: Develop a framework like MMDetection but *designed around selective scan*, with components such as anchor generation, NMS-free decoding, and spatial pyramid Mamba optimised for linear-complexity scanning. Target: 5 papers using the framework on canonical benchmarks.
- **Open-vocabulary Mamba detection**: Extend *Mamba-YOLO-World* (Wang et al. 2025a) (the only current entry in the corpus) into a systematic line of research. Integrate with vision-language models such as CLIP, SigLIP, or other foundation models. Target: $\geq$ 5 papers by the end of year 3.
- **Multi-modal canonical leaderboard**: A public leaderboard for RGB+depth, RGB+thermal, RGB+LiDAR detection with Mamba+YOLO as a backbone option. Extend existing benchmarks such as KITTI and nuScenes with Mamba-specific metrics (selective-scan efficiency, multi-modal token budget).

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10.1.4 Deliverables Overview

Table 18 summarizes the nine main deliverables together with their success metrics. To make the logical link between the synthesis and the roadmap explicit, the final column maps each deliverable to the specific challenge, opportunity, or risk it addresses from the 3×3 framework of Section 9.

Underpinning these applied deliverables are two cross-cutting *theoretical* directions that the roadmap deliberately leaves open. The first is the empirical characterisation

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Table 18: Three-Phase Future Research Roadmap for the Mamba+YOLO Community (36-Month Horizon), with each deliverable mapped to the Section 9 challenge (C), opportunity (O), or risk (R) it addresses.

Phase	Deliverable	Success metric	Addresses (§9)
I (0–12mo)	Benchmark Suite v1.0	$\geq$ 30 methods evaluated	Evaluation C; eval.-limits R
	Mamba Module Library	$\geq$ 100 GitHub stars/m	Software C (fragmentation)
	Code-release policy	$\geq$ 70% release rate	Reproducibility R
II (12–24mo)	Medical Mamba initiative	$\geq$ 20 medical papers	App.-gap R; underexpl. O
	Edge Deployment Toolkit	5 benchmarks on Hailo-8	Hardware C
	Foundation Mamba	3 ImageNet-22K checkpoints	Foundation-model O
III (24–36mo)	Mamba-native framework	5 papers using framework	Software C; hybrid O
	Open-vocab Mamba det.	$\geq$ 5 OV-Mamba papers	Multimodal/OV O
	Multi-modal leaderboard	3 datasets (KITTI/nuScenes)	Multimodal O; eval. R

of *state-space scaling laws* for detection—how detection accuracy and the selective-scan compute budget trade off as Mamba width, depth, and state dimension grow—which would give the field the parameter-count guidance that scaling studies already provide for CNN and Transformer detectors. The second is *hardware-aware selective scanning*: co-designing the scan order and chunk size with the memory hierarchy of edge accelerators (Jetson, Hailo-8, mobile NPUs), since the recurrent memory-access pattern of the selective scan, rather than its FLOPs, is the binding constraint on these platforms (Section 9.4). Both are prerequisites for moving Mamba+YOLO from accuracy-driven research to deployment-grade engineering.

Threats to Validity

We acknowledge the following threats to the validity of this SLR:

- **Selection bias.** Although other SLRs cover multiple databases, this review searched only Scopus, which may have missed the most recent conference papers indexed only on Web of Science or IEEE Xplore. Partially mitigated by Scopus's broad coverage of major computer-vision and engineering venues.
- **Retrieval bias.** The full-text retrieval rate was 66% (185/279), leaving 33.7% (94 papers) inaccessible and excluded under EC-5 (full text not retrievable). This share exceeds the 30% level at which retrieval bias becomes a material concern, so we report it explicitly as a notable limitation rather than treating the corpus as exhaustive. Institutional access at VSB-TUO mitigates but does not eliminate the issue.

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- **Subjectivity in quality assessment.** QA scoring combined automated heuristics with manual verification, which may introduce a ∼15–20% margin of error in borderline cases. Mitigated by the Phase 6 two-pass intra-rater scoring protocol (heuristic-based + criterion-based with ≥ 7-day separation) for borderline-PENDING entries. Inter-rater reliability was quantified by two independent co-authors: the second author (R. Jaroš) double-scored the full corpus (97.3% raw inclusion agreement, 108/111 independently re-scored), and—to avoid the high-agreement/low-κ paradox on that homogeneous accepted set—the third author (L.D. Minh) re-scored a full-range sample (sub-threshold to top-scored) for the κ estimate (quadratic-weighted Cohen's $\kappa = 0.86$; inclusion-decision $\kappa = 0.79$; Section 2.5), together indicating reproducible decisions.
- **Publication bias.** Negative results and null findings are under-represented in the published literature; this review cannot correct for this bias and we did not assess it systematically.
- **Geographic and benchmark-domain bias.** First-author affiliations are heavily concentrated in China (78%, Section 2.8) and the corpus is dominated by a few benchmark domains—remote sensing, autonomous driving, UAV/drone, and industrial defect—while medical/clinical and infrared/thermal applications remain sparse. Because the underlying benchmarks therefore over-represent particular regions, scene types, and capture conditions, the synthesised accuracy patterns and design preferences may not transfer uniformly to under-represented regions and domains. We accordingly read the geographic and domain shares as descriptive of where the field currently concentrates its effort, not as evidence of where Mamba+YOLO is most effective.
- **Data-extraction error.** The 16-field extraction relied on heuristic regex with manual verification; a residual error of ∼10% on numeric fields (mAP, FPS, Params, FLOPs) is plausible. The structured dataset is published openly to enable community-side correction. The categorical taxonomy fields (variant family, integration position, application domain) were likewise coded by the first author using regex-assisted heuristics with manual verification, without a separate inter-rater check on the classification itself—the κ analysis in Section 2.5 covers QA scoring and the inclusion decision, not category assignment. Borderline assignments, most visibly the nominal split between *Mamba (custom)* and *SSM (custom)* variants, are reported transparently and can be re-coded from the open dataset.
- **Time-window cutoff.** The review covers December 2023–May 2026. Studies published after May 2026 are not included; given the field's ∼10× year-on-year growth rate, the review should be read as a snapshot of the field's first ∼30 months.

Acknowledgements. The authors are deeply grateful to VSB – Technical University of Ostrava, Czech Republic, for the professional and comprehensive support provided to this research. The authors also thank the anonymous reviewers for their constructive feedback during the peer-review process.

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Declarations

Funding. This article was funded by the European Union under the REFRESH – Research Excellence For REgion Sustainability and High-tech Industries project number CZ.10.03.01/00/22_003/0000048 via the Operational Programme Just Transition. This article was funded by the Ministry of Education of the Czechia (Project No. SP2026/012, SP2026/075). The views and conclusions in this review are entirely those of the authors and do not necessarily reflect the views of the funding organizations.
Conflict of interest. The authors declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
Ethics approval and consent to participate. Not applicable. This study is a systematic literature review of previously published work and did not involve any human participants, human data, or animals.
Consent for publication. Not applicable.
Data availability. The structured 16-field extraction dataset for all 114 included studies is provided as supplementary material to support secondary research and replication; the Phase 6 adjudication record and the per-query overlap matrix are available from the corresponding author upon reasonable request. No other datasets were generated; all primary studies analysed in this review are listed in the reference list.
Materials availability. Not applicable.
Code availability. The scripts used to generate the figures and tables in this review are available from the authors upon reasonable request.
Author contributions. Thanh Kiet Vo: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization. Rene Jaros: Validation, Writing – review & editing. Ly Duc Minh: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization. Petr Bilik: Validation, Supervision, Project administration, Funding acquisition. Radek Martinek: Validation, Supervision, Project administration, Funding acquisition. All authors read and approved the final manuscript.
Use of generative AI in scientific writing. During the preparation of this manuscript the authors used a large language model to assist with language editing and translation from Vietnamese to English. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No content was generated and accepted unedited; all conclusions, arguments, taxonomy, empirical analyses, and findings are the authors' own.

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Appendix A Scopus Search Query Strings (Q1–Q6)

To support reproducibility of the identification stage, the full Boolean query strings applied to the Scopus database are listed below. Each query covered the publication window December 2023–May 2026, was filtered by document type *(article, conference paper, review)*, with no subject-area (SUBJAREA) restriction applied so as to maximise cross-disciplinary recall; the English-language requirement was enforced at the

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screening stage as inclusion criterion IC-3 rather than as a Scopus retrieval filter, and was executed on Scopus in two passes—an initial search in February 2026 retrieving the 2024–2025 records, and an update search on 6 May 2026 retrieving the 2026 records—which together cover the full window.

Q1 — Broad scope: Mamba + Detection

```
TITLE-ABS-KEY(("Mamba" OR "selective scan" OR "state space model" OR SSM)
AND ("object detection" OR YOLO))
```

Q2 — Mamba + YOLO precise

```
TITLE-ABS-KEY(("Mamba" OR "Vim" OR "VMamba" OR "SS2D") AND YOLO*)
```

Q3 — Vision Mamba

```
TITLE-ABS-KEY("Vision Mamba" OR "Vim" OR "VMamba") AND TITLE-ABS-KEY("object
detection" OR "detector")
```

Q4 — Selective scan + detection (anti-FP NLP)

```
TITLE-ABS-KEY("selective scan" AND ("detection" OR "detector")) AND NOT
TITLE-ABS-KEY("language model" OR NLP)
```

Q5 — SSM + YOLO

```
TITLE-ABS-KEY(("state space" OR S4 OR S6) AND YOLO* AND detection)
```

Q6 — RT-DETR + Mamba (Transformer-detector intersection)

```
TITLE-ABS-KEY((RT-DETR OR DETR) AND (Mamba OR "selective scan" OR "state
space"))
```

A total of 1,053 records were collected through these six queries (after merging and de-duplication, 522 records entered the screening stage). Per-query distribution: Q1 (412), Q2 (309), Q3 (147), Q4 (88), Q5 (62), Q6 (35); detailed overlap is available from the corresponding author upon request.

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Appendix B Glossary of Acronyms

Table B1 lists the acronyms used throughout this review.

Table B1: Glossary of Acronyms Used in This Review.

Acronym	Definition
AGI	Application Gap Index (Eq. 5)
AP	Average Precision
BFI	Bidirectional Fusion Integration
BM3D	Block-Matching 3D filtering
CFM	Cross Fusion Mamba (block)
CNN	Convolutional Neural Network
CSPVSS	Cross-Stage Partial Visual State Space
DETR	Detection Transformer
DFL	Distribution Focal Loss
ELAN	Efficient Layer Aggregation Network
EMA	Exponential Moving Average
FPN	Feature Pyramid Network
FPS	Frames Per Second
GAM	Global Attention Mechanism
HiPPO	High-order Polynomial Projection Operator
IC/EC	Inclusion / Exclusion Criteria (PRISMA)
IoU	Intersection over Union
ITS	Intelligent Transportation Systems
mAP	mean Average Precision
MSSM	Multi-directional Selective Scan Module
NMS	Non-Maximum Suppression
NPU	Neural Processing Unit
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PTQ	Post-Training Quantization
QA	Quality Assessment
RQ	Research Question
RT-DETR	Real-Time DETR
S4	Structured State-Space Sequence model
S6	Selective State-Space (Mamba core)
SLR	Systematic Literature Review
SS2D	2D Selective Scan (VMamba module)
SSD	State-Space Duality (Mamba-2)
SSM	State-Space Model
Vim	Vision Mamba
VMamba	Visual Mamba (with SS2D)
VSS	Visual State Space (block)
YOLO	You Only Look Once
ZOH	Zero-Order Hold (discretization rule)

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